
Bellman Calibration for V -Learning in Offline Reinforcement Learning

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Reliable long-horizon value prediction is difficult in offline reinforcement learning
2 because fitted value methods combine bootstrapping, function approximation, and
3 distribution shift, while standard guarantees often require Bellman completeness
4 or realizability. We introduce *Bellman calibration*, a weak reliability criterion
5 requiring that states assigned similar predicted values have average Bellman targets
6 that agree with those predictions. This criterion yields a scalar calibration error
7 for diagnosing systematic numerical miscalibration, which we estimate from off-
8 policy data using doubly robust Bellman target estimates. We then propose *Iterated*
9 *Bellman Calibration*, a model-agnostic post-hoc procedure that recalibrates any
10 learned value predictor by fitting a one-dimensional map of its original prediction,
11 with histogram and isotonic variants. We prove finite-sample guarantees showing
12 that Bellman calibration error is controlled at one-dimensional nonparametric rates
13 without Bellman completeness or value-function realizability. Our value-error
14 bounds separate statistical estimation, finite-iteration, and approximation errors,
15 clarifying when calibration improves value prediction and when its gains are limited
16 by the information in the original predictor or insufficient coverage.

17 1 Introduction

18 Many applications require predicting the long-term consequences of a decision policy in a sequential,
19 stochastic environment. We study this problem in a Markov decision process (MDP), where the
20 goal is to estimate the long-run value of a target policy π , possibly from data collected under a
21 different behavior policy. This off-policy value-prediction problem arises throughout RL and in many
22 application domains: clinicians anticipate long-term outcomes under proposed treatment rules [56];
23 online platforms estimate customer lifetime value and retention under alternative recommendation
24 strategies [36, 51, 72]; and economists study downstream effects of counterfactual programs [12, 46].
25 In these settings, systematic miscalibration can distort policy comparisons, misstate long-run effects,
26 and lead to poor decisions.

27 Modern predictors such as neural networks and gradient-boosted trees are often miscalibrated: their
28 numerical predictions may systematically differ from the quantities they are intended to estimate
29 because of misspecification, limited data, or distribution shift [4, 22, 41, 73]. For long-horizon value
30 prediction, this concern is amplified by bootstrapping: fitted value methods regress on targets that
31 themselves depend on previous value estimates, so finite-iteration error, approximation error, and off-
32 policy shift can propagate into the final value scale [1, 18, 21, 38, 52]. Existing guarantees typically
33 control these errors by imposing strong structure, such as Bellman completeness, realizability, linear
34 approximation, or finite state spaces [11, 37, 69, 70]. Min-max alternatives relax some aspects of
35 fitted value iteration but introduce their own challenges, including saddle-point optimization, rich
36 critic or dual classes, and strong coverage requirements [14, 27, 53, 69].

Table 1: Why RL value predictions can be informative but poorly calibrated.

Failure mode	Calibration correction
Temporal or distribution shift: value estimates reflect past rewards, policies, or environments rather than the current target setting [30].	Update the value scale using recent or gold-standard trajectories, while retaining signal from the historical model.
Off-policy fitting: a high-capacity value model trained on large behavior-policy data can be informative but mis-scaled for the target-policy state-action distribution [20].	Use the large off-policy sample to learn the value predictor, then recalibrate its scale on a smaller on-policy or target-policy reference sample.
Finite Bellman iteration: computational limits can force early stopping or a shorter effective horizon, mis-scaling values for the intended return target [38].	Re-align predictions to the intended horizon without rerunning the full Bellman iteration.
Approximation and optimization error: misspecified value classes, restricted critics, or unstable saddle-point training can yield biased value predictions [54].	Remap the biased but informative prediction to observed or Bellman-consistent values, rather than requiring exact specification.

Calibration is a classical reliability property: among units assigned the same predicted value, the corresponding outcomes should agree with that value on average. In population terms, a calibrated predictor satisfies $f(X) = E[Y \mid f(X)]$. This suggests an RL analogue. Informally, a value predictor is calibrated if, among states assigned the same predicted value, the average target-policy Bellman backup matches that value; that is, $v(S) = E_{\pi}\{R + \gamma v(S') \mid v(S)\}$. This is a weak but meaningful form of Bellman consistency: it constrains only the scalar prediction rather than the full state-to-value mapping, while still ruling out systematic under- or overestimation within level sets of the prediction.

Despite its central role in supervised learning, where predictions are calibrated against observed labels, empirical frequencies, or conditional means [23, 24, 33, 44, 66, 67, 73], calibration remains comparatively underexplored for long-horizon value prediction in RL. Table 1 highlights why this is problematic: value estimates can be highly informative yet numerically misaligned because of temporal shift, off-policy fitting, finite Bellman iteration, or approximation and optimization error. Unlike in supervised learning, however, the calibration target is not directly observed; long-run returns must be inferred from one-step transitions through Bellman structure. We therefore study post-hoc Bellman calibration: a restricted recalibration of a pretrained value predictor that corrects systematic scale errors without a new high-dimensional Bellman fit, with finite-sample guarantees that do not require realizability or Bellman completeness.

Contributions. Contributions. Contributions. We connect two largely separate literatures: post-hoc calibration for supervised prediction and long-horizon value learning in RL. Existing work gives rich methods for learning value functions, but much less theory for model-agnostic reliability assessment and post-hoc correction of already learned value predictors. Our contributions are:

1. We formalize **Bellman calibration**, a dynamic analogue of supervised calibration for infinite-horizon value prediction. Bellman consistency implies calibration, but calibration is weaker: predictions need only match average Bellman backups among states assigned similar values.
2. We propose ℓ^2 -**Bellman calibration error** as a scalar diagnostic for systematic value miscalibration and develop debiased off-policy estimators of this error. We also prove a **calibration-refinement** bound showing that value error decomposes into calibration error and the approximation error of the best predictor based only on the original model output.
3. We introduce **post-hoc Bellman calibration** via histogram binning and isotonic regression. The methods apply to any value predictor and use doubly robust Bellman targets for off-policy data.
4. We prove **finite-sample guarantees** for histogram calibration, controlling both ℓ^2 -Bellman calibration error and value error. The bounds require neither Bellman completeness nor realizability and separate statistical, auxiliary-model, and iteration errors.

70 1.1 Related work

71 **Post-hoc calibration.** Our work builds on post-hoc calibration for supervised prediction [4, 22, 41, 73, 74]; see Roth [45] for an overview. This literature calibrates static predictions against observed labels, empirical frequencies, or conditional means. We study long-horizon value predictions in MDPs, where returns are not directly observed and must be inferred through Bellman structure. We are also related to causal calibration for static treatment-effect prediction [60, 67], but our setting is dynamic and long-horizon, with calibration targets defined by value functions rather than one-step conditional effects. The closest work is [62], who use isotonic post-hoc calibration of Q -functions inside Double RL policy-value estimators [28, 29]. Their calibration step is used as an auxiliary-model correction for a global value estimator. We instead study calibrated value prediction directly and give finite-sample guarantees for both Bellman calibration error and value error. The same framework also applies to Q -function calibration by replacing importance-weighted FVI with unweighted FQE and treating each state-action pair as the state.

83 **Relation to Bellman learning and fine-tuning.** Bellman calibration complements methods that change value learning itself, including minimax or adversarial Bellman objectives [17, 53, 55, 69], conservative updates [2, 30], and structural assumptions such as linearity, low rank, representations, or discretization [9, 43, 48, 65]. It can be applied after any such method because it only needs scalar value predictions and a reference sample. Unlike offline-to-online transfer, critic/policy fine-tuning, or safe policy improvement [31, 32, 40, 47], Bellman calibration fixes the target policy, value model, representation, and Bellman learner, and fits only a one-dimensional post-hoc map to the target value scale. Thus it is an output-scale correction rather than RL fine-tuning, with finite-sample guarantees for calibration and value error. Related work on calibrated dynamics uncertainty, conformal sets, and distributional RL targets different objects [5, 13, 35, 50, 75].

93 2 Calibration for Value Functions

94 2.1 Setup and Notation

95 We consider an MDP with state space $\mathcal{S}$, discrete action space $\mathcal{A}$, initial-state distribution ρ for S_0 , transition kernel $P(s' | s, a)$, reward function $r_0(s, a)$, and discount factor $\gamma \in [0, 1)$. Data are collected under a behavior policy $b_0(a | s)$. Each calibration sample is an i.i.d. transition (S, A, R, S') , where $(S, A) \sim \rho \times b_0$, $S' \sim P(\cdot | S, A)$, and $R = r_0(S, A) + \varepsilon$ with $E[\varepsilon | S, A] = 0$. We observe a calibration sample $\mathcal{C}_n := \{(S_i, A_i, R_i, S'_i)\}_{i=1}^n$.

100 Let $\pi(a | s)$ be the target policy. The value function under π is $v_0(s) = E_\pi[\sum_{t=0}^{\infty} \gamma^t R_t | S_0 = s]$, where E_π denotes the law induced by $A_t \sim \pi(\cdot | S_t)$ and $S_{t+1} \sim P(\cdot | S_t, A_t)$.

102 For a state-action function f and state function v , define

$$(\pi f)(s) := \sum_{a \in \mathcal{A}} \pi(a | s) f(s, a), \quad (b_0 f)(s) := \sum_{a \in \mathcal{A}} b_0(a | s) f(s, a), \quad (Pv)(s, a) := E[v(S') | S = s, A = a].$$

103 Write

$$w_\pi := \pi/b_0, \quad r_{0,\pi} := \pi r_0, \quad P_\pi := \pi P, \quad \mathcal{T}_\pi v := r_{0,\pi} + \gamma P_\pi v$$

104 for the policy Bellman operator. The value function is the unique bounded fixed point [6, 7]

$$v_0 = \mathcal{T}_\pi v_0. \tag{1}$$

105 For a measure μ on $\mathcal{S}$, write

$$\|f\|_{2,\mu} := \left\{ \int f(s)^2 \mu(ds) \right\}^{1/2}.$$

106 Let $L^2(\rho)$ denote the space of square-integrable state functions under ρ . Let $\|f\|_{n,S,A}$ and $\|g\|_{n,S'}$ denote empirical L^2 norms over $\{(S_i, A_i)\}_{i=1}^n$ and $\{S'_i\}_{i=1}^n$, respectively. We write $\lesssim$ for inequalities up to absolute constants and $[B] := \{1, \dots, B\}$.

109 2.2 Bellman Calibration: A Minimal Requirement

110 Let $\hat{v}$ be an estimated value function for the target policy π , fit on data independent of the calibration sample $\mathcal{C}_n$. Our goal is not to replace the underlying value-learning procedure, but to impose a weak

post-hoc Bellman consistency requirement on its output. Throughout, we call the scalar $\hat{v}(S)$ the predicted value; in the supervised calibration literature, this same scalar is often called a prediction score.

A natural starting point is *statewise Bellman consistency*:

$$\hat{v}(S) \approx E_\pi[R + \gamma\hat{v}(S') \mid S].$$

Equivalently, the Bellman residual $R + \gamma\hat{v}(S') - \hat{v}(S)$ should have conditional mean zero given the full state. In high-dimensional or continuous state spaces, this is a strong requirement: checking or enforcing it requires estimating a state-dependent conditional mean and is closely tied to structural assumptions such as Bellman completeness [11].

We instead require Bellman consistency only after grouping states by their predicted value:

$$\hat{v}(S) \approx E_\pi[R + \gamma\hat{v}(S') \mid \hat{v}(S)].$$

Thus, predictions should agree on average with their Bellman targets within groups of states assigned the same predicted value $\hat{v}(S)$. This is the dynamic analogue of post-hoc calibration in supervised prediction, where predictions are required to be correct on average conditional on their own numerical output [22, 41, 74]. The benefit is that a high-dimensional Bellman moment is reduced to a one-dimensional conditional moment, enabling post-hoc correction without requiring the full fitted Bellman update to be correctly specified.

Bellman calibration. For any candidate value function v , define the Bellman-calibration map

$$\Gamma_0(v)(s) := E_\pi[R + \gamma v(S') \mid v(S) = v(s)]. \quad (2)$$

We say that v is *Bellman calibrated* if

$$v(S) = \Gamma_0(v)(S) \quad \text{almost surely,}$$

or equivalently,

$$E_\pi[R + \gamma v(S') - v(S) \mid v(S)] = 0 \quad \text{almost surely.}$$

A necessary but weak condition. The true value function v_0 is Bellman calibrated. The Bellman equation gives $v_0(S) = E_\pi[R + \gamma v_0(S') \mid S]$; conditioning on the coarser predicted value $v_0(S)$ yields

$$v_0(S) = E_\pi[R + \gamma v_0(S') \mid v_0(S)] = \Gamma_0(v_0)(S).$$

Thus Bellman consistency implies Bellman calibration. The converse need not hold: because calibration only conditions on the scalar predicted value $v(S)$, state-level errors can average out within predicted-value groups [24, 60]. Hence Bellman calibration is a minimal reliability requirement, not a guarantee of value accuracy.

Reward-model interpretation. Define the reward implied by a value predictor v as

$$r_{\pi,v}(s) := v(s) - \gamma P_\pi v(s).$$

This is the one-step reward under which v would satisfy the Bellman equation for policy π . Bellman calibration is equivalent to

$$E_\pi[R - r_{\pi,v}(S) \mid v(S)] = 0.$$

Thus, among states assigned the same predicted long-run value, the reward implied by v matches the true one-step reward on average. When $\gamma = 0$, this reduces to ordinary regression calibration [45].

2.3 A Calibration–Refinement Bound for Value Error

We next relate Bellman calibration to value prediction error. The key point is that calibration is a post-processing operation: it can correct systematic Bellman residuals among states with the same predicted value, but it cannot recover information lost by reducing the state S to $\hat{v}(S)$.

For a candidate value function v , define the ℓ^2 **Bellman calibration error**

$$\text{Cal}_{\ell^2}^2(v) := \|v - \Gamma_0(v)\|_{2,\rho}^2 = E_\rho \left[\{E_\pi[R + \gamma v(S') - v(S) \mid v(S)]\}^2 \right]. \quad (3)$$

This criterion is zero exactly when v is Bellman calibrated, and measures the remaining Bellman residuals conditional on the predicted value.

149 The value predictors generated by the original prediction $\hat{v}(S)$ form the class

$$\mathcal{V}_{\hat{v}} := \{g \circ \hat{v} : g : \mathbb{R} \rightarrow \mathbb{R} \text{ measurable}\}, \quad \Pi_{\hat{v}} := L^2(\rho)\text{-projection onto } \mathcal{V}_{\hat{v}}.$$

150 Define the projected transition by

$$P_{\pi, \hat{v}} := \Pi_{\hat{v}} P_{\pi}, \quad P_{\pi, \hat{v}} f(s) = E_{\pi}[f(S') \mid \hat{v}(S) = \hat{v}(s)].$$

151 Define the discounted projected concentrability by

$$\mathcal{C}_{\hat{v}, \gamma} := \left\{ (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \left\| \frac{d(\rho P_{\pi, \hat{v}}^t)}{d\rho} \right\|_{\infty}^{1/2} \right\}^2.$$

152 **A1 Projected concentrability.** $\mathcal{C}_{\hat{v}, \gamma} < \infty$.

153 When ρ is stationary under P_{π} , it is also stationary under $P_{\pi, \hat{v}}$ by Lemma 2, so $\mathcal{C}_{\hat{v}, \gamma} = 1$. More gen-
 154 erally, $\mathcal{C}_{\hat{v}, \gamma}$ measures distribution drift from ρ : it is small when projected rollouts remain comparable
 155 to ρ , and large when they concentrate where ρ has little mass.

156 **Theorem 1** (Calibration–refinement bound). *Under A1,*

$$\|\hat{v} - v_0\|_{2, \rho} \leq \frac{\sqrt{\mathcal{C}_{\hat{v}, \gamma}}}{1 - \gamma} \left\{ \text{Cal}_{\ell^2}(\hat{v}) + \min_{\theta: \mathbb{R} \rightarrow \mathbb{R}} \|\theta \circ \hat{v} - v_0\|_{2, \rho} \right\}.$$

157 Theorem 1 gives a post-processing guarantee: value error is controlled by Bellman calibration error
 158 $\text{Cal}_{\ell^2}(\hat{v})$ plus the refinement error $\min_{\theta: \mathbb{R} \rightarrow \mathbb{R}} \|\theta \circ \hat{v} - v_0\|_{2, \rho}$, the error of the best value predictor
 159 that depends on the state only through $\hat{v}(S)$. The calibration error measures systematic Bellman
 160 residuals within levels of $\hat{v}(S)$, whereas the refinement error measures the remaining value variation
 161 not captured by $\hat{v}(S)$. Thus, calibration can remove Bellman miscalibration using only a post-
 162 processing map of the original prediction, but it cannot recover information about $v_0(S)$ absent
 163 from $\hat{v}(S)$. In particular, when calibration error is driven to zero, value error is controlled, up to
 164 the projected concentrability and discount factors, by the refinement error. This mirrors classical
 165 calibration–refinement decompositions in classification and regression [15, 39, 60, 67]. The factor
 166 $\sqrt{\mathcal{C}_{\hat{v}, \gamma}}/(1 - \gamma)$ plays the same role as the concentrability and discount penalties in classical FVI/FQE
 167 error bounds [38].

168 2.4 Evaluating Bellman Calibration Error with Doubly Robust Bellman Targets

169 We now focus on estimating $\text{Cal}_{\ell^2}(\hat{v})$, the component of Theorem 1 that can be evaluated from
 170 transition data and improved without refitting the value predictor. This makes Bellman calibration
 171 error both a diagnostic for systematic Bellman residuals within the level sets of a fixed predictor $\hat{v}$
 172 and an objective for post-hoc calibration.

173 Because data are generated under the behavior policy b_0 , we estimate the statewise Bellman target
 174 using a doubly robust Bellman pseudo-outcome, extending the calibration target estimates of [60, 71]
 175 to dynamic settings. Let $\hat{w}_{\pi}$ estimate $w_{\pi} := \pi/b_0$. For a fixed v , define

$$q_v(s, a) := E\{R + \gamma v(S') \mid S = s, A = a\}.$$

176 Let $\hat{q}_v$ estimate q_v . For $O = (S, A, R, S')$, set

$$\hat{Y}_v(O) := (\pi \hat{q}_v)(S) + \hat{w}_{\pi}(A \mid S) \{R + \gamma v(S') - \hat{q}_v(S, A)\}, \quad \hat{\mathcal{T}}_{\pi} v(s) := E\{\hat{Y}_v(O) \mid S = s\}.$$

177 The next result gives the conditional bias of this pseudo-outcome.

178 **Theorem 2** (Doubly robust Bellman target). *For any fixed v ,*

$$E\left[\{\hat{\mathcal{T}}_{\pi} v - \mathcal{T}_{\pi} v\}(S) \mid v(S) = t\right] = E[b_0\{(w_{\pi} - \hat{w}_{\pi})(\hat{q}_v - q_v)\}(S) \mid v(S) = t].$$

179 Consequently, $E\{\hat{Y}_v(O) \mid v(S)\} = E\{(\mathcal{T}_{\pi} v)(S) \mid v(S)\}$ if either $\hat{w}_{\pi} = w_{\pi}$ or $\hat{q}_v = q_v$.

180 This construction includes importance weighting ($\hat{q}_v = 0$) and model-based Bellman-target evaluation
 181 ($\hat{w}_\pi = 0$) as special cases. The theorem shows that either nuisance estimator suffices to recover the
 182 prediction-conditional Bellman target.

183 To estimate the proposed calibration error, we use a **cross-fitted debiased moment estimator**.
 184 Let $U_i := v(S_i)$ and $\hat{Y}_i := \hat{Y}_v(O_i)$. Following Xu and Yadlowsky [71], we estimate the squared
 185 calibration error by

$$\widehat{\text{Cal}}_{\ell^2, \text{db}}^2(v) := \frac{1}{n} \sum_{i=1}^n \{\hat{Y}_i - U_i\} \{\hat{m}_{-i}(U_i) - U_i\},$$

186 where $\hat{m}_{-i}$ is fit without observation i to estimate $u \mapsto E(\hat{Y}_v \mid v(S) = u)$. The population identity
 187 behind this estimator is

$$E\left[\{\hat{Y}_v - v(S)\} \left\{E(\hat{Y}_v \mid v(S)) - v(S)\right\}\right] = E\left[\left\{E(\hat{Y}_v \mid v(S)) - v(S)\right\}^2\right].$$

188 By Theorem 2, this equals $\text{Cal}_{\ell^2}^2(v)$, up to function approximation error.

189 3 Post-hoc Bellman Calibration

190 3.1 Iterated Bellman Calibration

191 Algorithm 1 gives a post-hoc FVI-style procedure that transforms an initial value predictor $\hat{v}$ into
 192 a calibrated predictor $V_{\text{cal}} = V_K = g_K \circ \hat{v}$. Starting from $V_0 = \hat{v}$, each iteration forms doubly
 193 robust Bellman targets $\hat{Y}_{V_k}(O_i)$ and regresses them on the fixed one-dimensional inputs $\hat{v}(S_i)$ over
 194 a calibrator class $\mathcal{G}$, yielding $V_{k+1} = g_{k+1} \circ \hat{v}$. Equivalently, for $\mathcal{V}_{\hat{v}}(\mathcal{G}) := \{g \circ \hat{v} : g \in \mathcal{G}\}$, the
 195 algorithm is a projected Bellman update restricted to $\mathcal{V}_{\hat{v}}(\mathcal{G})$. This avoids high-dimensional function
 196 approximation while correcting Bellman miscalibration in the original predictor; the contraction
 197 results below give geometric convergence, so $O\{\log(1/\varepsilon)/(1 - \gamma)\}$ iterations suffice to reach
 198 optimization error ε .

Algorithm 1 Iterated Bellman Calibration

- 1: **Input:** $\hat{v}, \mathcal{C}_n = \{O_i = (S_i, A_i, R_i, S'_i)\}_{i=1}^n$, Pseudo-outcome $v \mapsto \hat{Y}_v$, class $\mathcal{G}$, iterations K
 - 2: Initialize $V_0 = \hat{v}$
 - 3: **for** $k = 0, \dots, K - 1$ **do**
 - 4: $V_{k+1} = g_{k+1} \circ \hat{v}$ where $g_{k+1} \in \arg \min_{g \in \mathcal{G}} \sum_{i=1}^n \{\hat{Y}_{V_k}(O_i) - g(\hat{v}(S_i))\}^2$
 - 5: **end for**
 - 6: **Output:** V_K
-

199 Heuristically, V_K becomes calibrated because each iteration approximates one application of the
 200 projected Bellman operator $\mathcal{B}_{\hat{v}} f := \Pi_{\hat{v}} r_{0,\pi} + \gamma P_{\pi, \hat{v}} f$, whose fixed point over functions of $\hat{v}(S)$
 201 has zero Bellman calibration error. Exact finite-sample recovery of this projected fixed point would
 202 generally require realizability in $\mathcal{V}_{\hat{v}}(\mathcal{G})$ and Bellman completeness of the fitted update [38]. We
 203 instead seek guarantees for post-hoc improvement through Bellman calibration without imposing
 204 these conditions. We therefore instantiate Algorithm 1 with one-dimensional calibrators based on
 205 histogram regression [49] and isotonic regression [3]. These are dynamic analogues of histogram
 206 binning [23, 73, 74] and isotonic calibration [41, 57, 60, 74], and they admit finite-sample Bellman
 207 calibration guarantees without Bellman completeness.

208 3.2 Histogram Calibration

209 Histogram calibration instantiates Algorithm 1 with a piecewise-constant calibrator. Partition the
 210 initial values $\{\hat{v}(S_i)\}_{i=1}^n$ into ordered bins $I_1, \dots, I_B$, using either empirical quantiles (*equal-mass*)
 211 or uniform discretization (*equal-width*). We take

$$\mathcal{G}_B := \text{span}\{\mathbf{1}_{I_b} : b \in [B]\}, \quad \mathcal{H}_B(\hat{v}) := \{g \circ \hat{v} : g \in \mathcal{G}_B\},$$

212 so every $h \in \mathcal{H}_B(\hat{v})$ is a value predictor that is constant within bins of the original prediction. The
 213 bins may be data-adaptive, but B is deterministic and may grow with n .

214 With this class, each iteration simply averages the fitted Bellman targets within each bin:

$$g_{k+1}(t) = \frac{1}{n_b} \sum_{i: \hat{v}(S_i) \in I_b} \hat{Y}_i^{(k)}, \quad t \in I_b,$$

215 where $n_b := \sum_{i=1}^n \mathbf{1}\{\hat{v}(S_i) \in I_b\}$. Thus, histogram calibration is fitted Bellman iteration on a finite
 216 aggregation of the initial predicted value. At convergence, the output approximately satisfies the
 217 empirical aggregated Bellman equation within each bin:

$$V_K(s) \approx E_n[\hat{Y}_{V_K}(O) \mid \hat{v}(S) \in I_b], \quad \hat{v}(s) \in I_b.$$

218 This is the empirical analogue of Bellman calibration and is the key property used in the next result.

219 **Calibration error.** We next give finite-sample, finite-iteration bounds for the Bellman calibration
 220 error (3). The analysis uses the following regularity conditions.

221 **C1 Boundedness.** $R, w_\pi, \hat{w}_\pi, \hat{v}$, the Bellman targets q_f , and their estimates $\hat{q}_f$ are uniformly
 222 bounded by $M < \infty$ for all f considered.

223 **C2 Sample splitting.** The auxiliary estimators $\hat{w}_\pi$ and $v \mapsto \hat{q}_v$, and the initial predictor $\hat{v}$ are trained
 224 on data independent of $\mathcal{C}_n$.

225 **C3 Stability of the Bellman-target regression.** For all $f, g \in \mathcal{H}_B(\hat{v})$ with $\|f\|_\infty, \|g\|_\infty \leq M$,

$$\|\hat{q}_f - \hat{q}_g\|_{n,S,A} \leq L \|f - g\|_{n,S'}.$$

226 These conditions are mainly technical. Boundedness can be enforced by clipping or truncation, and
 227 sample splitting can be replaced by cross-fitting. Condition C3 controls the complexity introduced by
 228 the map $v \mapsto \hat{q}_v$; it holds trivially for the importance-weighted target ($\hat{q}_v = 0$), and holds with $L = 1$
 229 for cellwise estimates of $E\{R + \gamma f(S') \mid S, A\}$.

230 **Theorem 3** (Calibration error for histogram binning). *Assume C1–C3. Then there is a constant*
 231 *$C < \infty$ such that, for any $K \in \mathbb{N}$, with probability at least $1 - \delta$,*

$$\begin{aligned} \text{Cal}_{\ell^2}(V_K) \leq C & \left\{ \sqrt{\frac{B}{n} \log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(1/\delta)}{n}} \right\} \\ & + \|b_0\{(\hat{w}_\pi - w_\pi)(\hat{q}_{V_K} - q_{V_K})\}\|_{2,\rho} + \|\hat{\mathcal{T}}_\pi V_K - \hat{\mathcal{T}}_\pi V_{K-1}\|_{2,\rho}. \end{aligned}$$

232 The bound separates three errors. The first line is the statistical error of one-dimensional histogram
 233 calibration, matching classical rates when the calibration target is observed directly [23, 24, 57]. The
 234 second line is the doubly robust auxiliary-model error from Theorem 2. The third line is the finite-
 235 iteration error, which decays geometrically as γ^K under the contraction conditions in Appendix K.4,
 236 as in fitted value iteration [38].

237 The leading calibration rate does not accumulate over the horizon: γ enters only through the finite-
 238 iteration and auxiliary-model terms. This reflects that Bellman calibration is a one-step conditional
 239 prediction problem, even though the value function itself is long-horizon. However, Theorem 1 shows
 240 that translating calibration into value accuracy still depends on discounting and coverage.

241 **Value error.** We next show that post-hoc calibration does not add error beyond the information loss
 242 from restricting to the histogram calibrator class, while it can remove Bellman miscalibration in the
 243 original predictor.

244 Let Π_B denote the $L^2(\rho)$ -projection onto $\mathcal{H}_B(\hat{v})$, i.e.,

$$\Pi_B q := \arg \min_{h \in \mathcal{H}_B(\hat{v})} \|q - h\|_{2,\rho}, \quad \mathcal{B}_{\pi,B} := \Pi_B \mathcal{T}_\pi, \quad V_B^* = \mathcal{B}_{\pi,B} V_B^*.$$

245 Write $P_{\pi,B} := \Pi_B P_\pi$, so that, for $\hat{b}(s) = b$ when $\hat{v}(s) \in I_b$,

$$P_{\pi,B} f(s) = E_\pi[f(S') \mid \hat{v}(S) \in I_{\hat{b}(s)}].$$

246 Finally, define the aggregated discounted concentrability by

$$\mathcal{C}_{B,\gamma} := \left\{ (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \left\| \frac{d(\rho P_{\pi,B}^t)}{d\rho} \right\|_\infty^{1/2} \right\}^2.$$

247 **C4 Aggregated projected concentrability.** $\mathcal{C}_{B,\gamma} < \infty$.

248 **Theorem 4** (Value error for histogram calibration). Assume C1–C4. Then there exists a constant
 249 $C < \infty$ such that, for any $K \in \mathbb{N}$, with probability at least $1 - \delta$,

$$\begin{aligned} \|V_K - v_0\|_{2,\rho} &\leq \gamma^K \|\hat{v} - V_B^*\|_\infty + \frac{\sqrt{\mathcal{C}_{B,\gamma}}}{1 - \gamma} \min_{h \in \mathcal{H}_B(\hat{v})} \|h - v_0\|_{2,\rho} \\ &\quad + C \frac{\sqrt{\mathcal{C}_{B,\gamma}}}{1 - \gamma} \left[\sqrt{\frac{B}{n} \log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(K/\delta)}{n}} + \max_{0 \leq j \leq K} \|b_0\{(\hat{w}_\pi - w_\pi)(\hat{q}_{V_j} - q_{V_j})\}\|_{2,\rho} \right]. \end{aligned}$$

250 **Proof idea.** The result avoids realizability and Bellman completeness for the original value class
 251 because completeness holds after aggregation. For any $f \in \mathcal{H}_B(\hat{v})$, the aggregated transition $P_{\pi,B}f$
 252 is binwise constant in $\hat{v}$, so $P_{\pi,B}f \in \mathcal{H}_B(\hat{v})$ and hence $\mathcal{B}_{\pi,B}f \in \mathcal{H}_B(\hat{v})$. The proof uses sup-norm
 253 contraction to obtain the aggregated fixed point, then propagates ρ -norm one-step regression errors
 254 through a discounted resolvent controlled by $\mathcal{C}_{B,\gamma}$. It also bounds $V_B^* - v_0$ by a histogram analogue
 255 of Theorem 1. Unlike prior discretization arguments for Bellman consistency [11, 43, 65, 68, 69], we
 256 discretize $\hat{v}(S)$ to target Bellman calibration.

257 **Discussion.** Theorems 3 and 4 separate calibration from value accuracy. With known importance
 258 weights and negligible iteration error, histogram calibration drives Bellman calibration error to zero at
 259 the one-dimensional rate $\sqrt{(B/n) \log(n/B)}$. Value error contains the same stochastic term, but also
 260 includes the prediction-only approximation error $(\sqrt{\mathcal{C}_{B,\gamma}})/(1 - \gamma) \min_{h \in \mathcal{H}_B(\hat{v})} \|h - v_0\|_{2,\rho}$, the
 261 cost of restricting the calibrated predictor to binned transformations of the original prediction. Thus
 262 B controls the usual bias–variance tradeoff: coarse bins may discard information in $\hat{v}$, while fine bins
 263 increase variance. The coverage factor $\mathcal{C}_{B,\gamma}$ is the histogram analogue of FVI concentrability.

264 It suffices to choose B by predictive cross-validation. The prediction-only approximation term
 265 satisfies

$$\min_{h \in \mathcal{H}_B(\hat{v})} \|h - v_0\|_{2,\rho} \leq \|\hat{v} - v_0\|_{2,\rho} + \min_{h \in \mathcal{H}_B(\hat{v})} \|h - \hat{v}\|_{2,\rho}.$$

266 Thus binning is no worse than the original predictor, up to identity approximation on the range of
 267 $\hat{v}(S)$. For equal-mass bins, the histogram bias–variance balance suggests $B \asymp n^{1/3}$, yielding [25]

$$\|V_K - v_0\|_{2,\rho} \leq \frac{\sqrt{\mathcal{C}_{B,\gamma}}}{1 - \gamma} \|\hat{v} - v_0\|_{2,\rho} + O_p\{(\log n/n)^{1/3}\},$$

268 with Bellman calibration error vanishing at the same rate.

269 3.3 Isotonic Calibration

270 Histogram calibration requires choosing the number of bins B . As a low-tuning alternative, we
 271 consider *isotonic iterated Bellman calibration*, obtained by taking $\mathcal{G} = \mathcal{G}_{\text{iso}}$ in Algorithm 1, where
 272 $\mathcal{G}_{\text{iso}}$ is the class of nondecreasing functions on $\mathbb{R}$. Each update isotonic-regresses the fitted Bellman
 273 targets on the initial values of $\hat{v}(S_i)$, yielding a monotone calibrator. This can be computed in
 274 near-linear time by the pool-adjacent-violators algorithm (PAVA) [8].

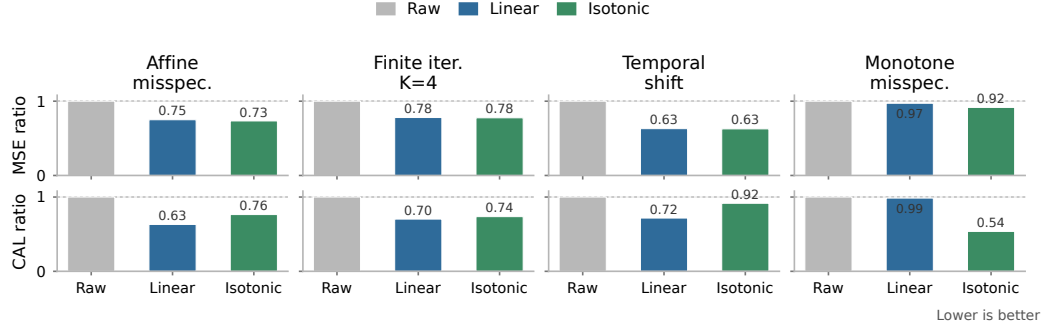
275 Like histogram calibration, isotonic calibration averages Bellman targets over groups of nearby
 276 predicted values, but the partition is chosen adaptively by the monotonicity constraint [3]. This
 277 provides regularization without selecting B . Since the identity map is monotone, isotonic calibration
 278 can preserve an already reliable predictor while correcting systematic monotone distortions.

279 **C5 Finite variation.** There exists $V < \infty$ such that, almost surely, the function $t \mapsto$
 280 $E[(\mathcal{T}_\pi V_K)(S) \mid \hat{v}(S) = t, \mathcal{C}_n]$ has total variation at most V .

281 Condition C5 is standard in isotonic calibration [60] and controls the complexity of the one-
 282 dimensional calibration curve. For histogram binning, the analogous complexity is governed by the
 283 number of bins B .

284 **Theorem 5** (Calibration error for isotonic calibration). Assume C1, C2, C3, and C5 with $\mathcal{G} = \mathcal{G}_{\text{iso}}$.
 285 Then there exists a constant $C < \infty$ such that, for any $K \in \mathbb{N}$, with probability at least $1 - \delta$,

$$\text{Cal}_{\ell_2}(V_K) \leq C \left\{ n^{-1/3} + \sqrt{\frac{\log(1/\delta)}{n}} \right\} + \|b_0\{(\hat{w}_\pi - w_\pi)(\hat{q}_{V_K} - q_{V_K})\}\|_{2,\rho} + \|\hat{\mathcal{T}}_\pi V_K - \hat{\mathcal{T}}_\pi V_{K-1}\|_{2,\rho}.$$



(a) Synthetic failure modes.

(b) **Post-hoc value calibration improves mis-scaled value predictions.** Value MSE and Bellman calibration error ratios across synthetic failure modes, relative to the matched uncalibrated baseline.

286 The isotonic analysis parallels the histogram case, but uses the adaptive pooling induced by the
 287 pool-adjacent-violators algorithm. We do not assume the calibration target is monotone; monotonicity
 288 is used only as a regularizer for pooling nearby predicted values. The resulting $n^{-1/3}$ rate matches
 289 the classical isotonic regression rate [10] and the rate of histogram binning with $B \asymp n^{1/3}$, but avoids
 290 choosing B in advance.

291 Our histogram proof does not yield value-error bounds for fully iterated isotonic calibration without
 292 Bellman completeness, because the PAVA partition can change across Bellman iterations. Ap-
 293 pendix D.1 gives a hybrid procedure that first learns the partition by isotonic regression and then
 294 applies the histogram analysis conditionally on that fixed partition.

295 4 Experimental investigation

296 We evaluate Bellman calibration as a post-hoc correction for fitted off-policy value predictors. Each
 297 base learner estimates a Q -function and forms the target-policy value predictor $\hat{V}$; calibration then
 298 fits a one-dimensional map g , using importance-weighted fitted value iteration, and reports $g \circ \hat{V}$.
 299 We use linear FQE and random-feature FQE as base learners, with linear, histogram, and isotonic
 300 calibrators. The experiments isolate the failure mode in Table 1: $\hat{V}$ is informative but mis-scaled. For
 301 non-temporal-shift settings, we use 5-fold cross-calibration [60]; for temporal shift, we train FQE on
 302 a large old-regime batch, fit g on a small current-regime batch, and evaluate on current-regime states.
 303 All main results use 100 independent replications and independent evaluation data. Values below one
 304 indicate improvement. See Appendix C for details.

305 Figure 1a and Table 2 in Appendix A show consistent gains across four failure modes. Under affine
 306 misspecification, linear calibration reduces true- V MSE by 25% and Bellman calibration error by
 307 37%. With finite Bellman-iteration error, random-feature FQE predictions remain rank-informative
 308 but under-scaled; linear and isotonic calibration reduce true- V MSE by 22% and Bellman calibration
 309 error by 26–34%. Under temporal reward shift, recalibrating an old-regime FQE model on a small
 310 current-regime batch reduces current-regime MSE by 37% and Bellman calibration error by 28%,
 311 outperforming a same-size retrain. Richer calibrators help under monotone misspecification: isotonic
 312 and histogram calibration nearly halve Bellman calibration error, while linear calibration has only
 313 mild effect. Overall, linear calibration corrects affine scale errors, whereas isotonic calibration handles
 314 more general monotone distortions with no tuning.

315 **Conclusion.** Bellman calibration provides an estimable, post-hoc reliability criterion for long-
 316 horizon value prediction: predictions should agree, on average, with their target-policy Bellman
 317 backups among states assigned similar values. Iterated Bellman Calibration turns this testable
 318 criterion into a one-dimensional recalibration procedure that can control calibration error without
 319 Bellman completeness or value-function realizability. Appendix B discusses the main takeaways,
 320 practical limitations from coverage and nuisance estimation, and future directions including stationary
 321 weighting, smoother calibrators, and larger offline RL evaluations.

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A Table for experiment

Table 2: **Value-scale correction across offline RL failure modes.** Ratios are relative to the matched uncalibrated baseline. Win rate is the seed-matched true- V MSE win rate.

Regime and learner	Calibrator	Rel. V MSE	Rel. cal. err.	Win rate
Affine misspec., linear FQE	Linear	0.750	0.631	0.94
Affine misspec., linear FQE	Isotonic	0.734	0.764	0.94
Finite iteration, RF FQE $K = 4$	Linear	0.780	0.703	0.94
Finite iteration, RF FQE $K = 4$	Isotonic	0.776	0.736	0.94
Temporal reward shift, RF FQE	Linear recent	0.630	0.717	0.97
Temporal reward shift, RF FQE	Isotonic recent	0.627	0.916	0.98
Monotone misspec., linear FQE	Linear	0.972	0.986	0.64
Monotone misspec., linear FQE	Isotonic	0.916	0.537	0.79
Monotone misspec., linear FQE	Histogram	0.917	0.544	0.77

B Conclusion and Future Work

We introduced Bellman calibration as a post-hoc reliability criterion for long-horizon value prediction. The criterion asks that predictions agree, on average, with their target-policy Bellman backups within groups of states assigned similar predicted values. This is weaker than statewise Bellman consistency, but strong enough to diagnose systematic value-scale errors and to support model-agnostic recalibration of an already learned value predictor.

Iterated Bellman Calibration turns this criterion into a practical correction: fit a one-dimensional calibration map using off-policy Bellman targets, then iterate the induced calibrated Bellman update. Because the recalibration is one-dimensional, histogram and isotonic variants can achieve finite-sample control of Bellman calibration error without Bellman completeness or value-function realizability. The resulting value-error bounds separate statistical calibration error, finite-iteration error, nuisance-estimation error, and the approximation error of the best predictor expressible as a function of the original value estimate. Thus calibration is most useful when the base predictor is informative but mis-scaled, and its limits are explicit when the original prediction discards information needed for accurate value prediction.

The main practical challenges are coverage, nuisance estimation, and extrapolation. Off-policy Bellman calibration requires enough overlap to estimate target-policy Bellman backups on the calibration sample, and poor coverage can make importance-weighted or doubly robust targets unstable. The method also corrects the value scale through the original predictor, so one-dimensional calibration cannot recover state information absent from that predictor. Future work includes combining Bellman calibration with stationary-density-ratio or emphatic-style weighting [34, 42, 59], developing smoother calibrators for extrapolation under distribution shift [22, 26, 44], and studying calibration as both a diagnostic and post-hoc correction in larger offline RL benchmarks.

C Additional Experimental Details

For the non-temporal-shift experiments, we use 5-fold cross-calibration [60]. In each fold, the value learner and calibrator are trained out of fold, and calibrated predictions are evaluated on the held-out fold. Fold-specific calibrated predictors are aggregated by pointwise medians. The uncalibrated comparator uses the same base learner refit on the full training sample. Calibration is fit by importance-weighted fitted value iteration, using the target-to-behavior action-probability ratio as weights. In the temporal-shift experiment, we train FQE on a large old reward-regime batch, fit the calibration map on a small held-out current-regime batch, and evaluate only on current-regime states. All main results use 100 independent replications and independent evaluation data.

Data-generating processes. All four simulation panels use discounted MDPs with continuous states $S_t \in \mathbb{R}^d$, actions $A_t \in \{1, 2, 3\}$, and softmax behavior and target policies. The target policy has logits $s^\top w_a^\pi / \tau_\pi$. The behavior policy uses shifted logits $s^\top w_a^b / \tau_b$, where $w_a^b = w_a^\pi - \Delta u_a$. The main experiments use the moderate-shift setting. Initial states are Gaussian, $S_0 \sim N(\mu_0, I_d)$. Except where noted below, transitions and rewards are

$$S_{t+1} = \tanh\{MS_t + \alpha_{A_t} + 0.22 \sin(S_t) + 0.08 \cos(S_t^{\text{rev}})\} + \sigma_\varepsilon \varepsilon_t,$$

and

$$r_0(s, a) = s^\top \beta + 0.65 \sin(s_1) + 0.35 \cos(s_2) + s^\top \eta_a + \rho_a - 0.06 \|s\|_2^2.$$

Observed rewards add mean-zero Gaussian noise. The ground-truth value $V^\pi(s)$ is computed by independent target-policy Monte Carlo rollouts using the conditional mean reward. Training transitions, calibration folds, Bellman-evaluation transitions, initial evaluation states, and Monte Carlo rollouts use independent seeds.

The four panels isolate different value-scale errors. In the affine misspecification panel, the same nonlinear MDP is used, but the reward is transformed to

$$r_{\text{aff}}(s, a) = 1.25 + 0.7 r_0(s, a).$$

The learner is linear FQE with a deliberately restricted feature class. This produces informative value predictions with an affine slope/intercept error, so linear calibration should be sufficient.

In the finite-iteration panel, the dynamics are made persistent,

$$S_{t+1} = 0.92 S_t + \alpha_{A_t} + 0.03 \sin(S_t) + \sigma_\varepsilon \varepsilon_t,$$

with $\gamma = 0.985$ and horizon 120. Rewards are smooth but long-horizon:

$$r_{\text{fin}}(s, a) = 0.8 + 0.9 \tanh(s_1) + 0.25 s_2 + \rho_a + 0.15 \tanh(s_1 + 0.5 s_2) + s^\top \eta_a.$$

Random-feature FQE is stopped after $K = 4$ Bellman iterations, so its values retain useful ranking information but are under-scaled relative to the long-horizon target.

In the temporal reward-shift panel, the old and current environments share dynamics, policies, and state distribution. Only the reward changes:

$$r_{\text{current}}(s, a) = 2.0 + 1.5 r_{\text{old}}(s, a).$$

We train random-feature FQE on a large old-regime behavior batch ($n_{\text{old}} = 2000$), fit the calibration map on a small recent current-regime batch ($n_{\text{recent}} = 100$), and evaluate on independent current-regime ground-truth values and Bellman-calibration transitions. A same-size current-regime retrain comparator uses the same learner.

In the monotone-misspecification panel, the nonlinear reward is transformed by a monotone saturation,

$$r_{\text{mono}}(s, a) = \text{sign}\{r_0(s, a)\} \sqrt{|r_0(s, a)| + 1}.$$

The linear FQE value prediction remains sufficiently ordered to be useful but has a nonlinear value-scale distortion. This is the setting where isotonic and histogram calibration should improve over an affine correction.

Baseline value predictors. Each method first fits a Q -function from the offline behavior-policy data and then forms

$$\hat{V}(s) = \sum_a \pi(a | s) \hat{Q}(s, a).$$

The experiments use random-feature FQE and linear FQE because they give transparent error mechanisms: finite Bellman-iteration bias and affine or monotone value-scale misspecification.

Cross-calibration. For fold k , the Q -learner is fit on the other folds. The held-out fold supplies $(\hat{V}^{(-k)}(S_i), \hat{V}^{(-k)}(S'_i), R_i)$. These out-of-fold tuples are pooled to fit a single value-space calibration map by importance-weighted fitted value iteration. At evaluation time, calibrated values are aggregated as

$$\hat{V}_{\text{cal}}(s) = \text{median}_{k=1, \dots, K_{\text{fold}}} g\left(\hat{V}^{(-k)}(s)\right).$$

The matched uncalibrated comparator is a Q -learner refit on all available training data.

Algorithm 2 Iso–Hist Iterated Bellman Calibration

- 1: **Input:** Value predictor $\hat{v}$, calibration data $\mathcal{C}_n$, Bellman target-estimate map $v \mapsto \hat{Y}_v$, iterations K
 - 2: **Stage 1: fit outcome-adaptive partition**
 - 3: $\hat{Y}_i^{(0)} \leftarrow \hat{Y}_{\hat{v}}(O_i)$
 - 4: $g_{\text{iso}} \leftarrow \arg \min_{g \in \mathcal{G}_{\text{iso}}} \sum_{i=1}^n (\hat{Y}_i^{(0)} - g(\hat{v}(S_i)))^2$
 - 5: Extract bins $\hat{I}_1, \dots, \hat{I}_{\hat{B}}$ from flat regions of g_{iso}
 - 6: Define $\hat{\mathcal{G}}_{\hat{B}} := \{g : g \text{ is constant on each } \hat{I}_b\}$
 - 7: **Stage 2: histogram calibration**
 - 8: Apply Algorithm 1 with class $\hat{\mathcal{G}}_{\hat{B}}$ to obtain V_K
 - 9: **Output:** V_K
-

582 **Value-space fitted value iteration.** Let $\omega_i = \pi(A_i \mid S_i)/b(A_i \mid S_i)$. In all reported runs the
 583 action-ratio weights are clipped at 20 and normalized to mean one within each calibration fit. Starting
 584 from the identity map, calibration iterates

$$g_{t+1} \in \arg \min_g \sum_i \omega_i \left[g(\hat{V}(S_i)) - \{R_i + \gamma g_t(\hat{V}(S'_i))\} \right]^2.$$

585 The implemented calibrators are weighted affine regression, weighted histogram binning, weighted
 586 isotonic regression, and a weighted isotonic-histogram hybrid.

587 **Metrics.** The scalar OPE estimate is the mean of $g(\hat{V}(S_0))$ over independent initial states. The
 588 main prediction metric is held-out value-function MSE,

$$\mathbb{E} \left[\{g(\hat{V}(S)) - V^\pi(S)\}^2 \right],$$

589 estimated using independent target-policy ground-truth values. Bellman calibration error is estimated
 590 on independent Bellman-evaluation transitions by binning $g(\hat{V}(S))$ into quantile bins and averaging
 591 the weighted squared gaps between bin-mean predictions and bin-mean Bellman outcomes

$$r_0(S, A) + \gamma g(\hat{V}(S')).$$

592 We use independent evaluation batches of size 50,000 and 50 bins.

593 D Additional details and extensions

594 D.1 Hybrid Isotonic–Histogram Calibration

595 Isotonic iterated Bellman calibration adaptively pools nearby predicted values through the monotonic-
 596 ity constraint. This adaptivity is useful in practice, but it complicates the estimation-error analysis
 597 because applying Algorithm 1 with $\mathcal{G} = \mathcal{G}_{\text{iso}}$ may produce a different data-dependent partition at
 598 each iteration. The iterates therefore need not correspond to a single aggregated Bellman operator.

599 The hybrid isotonic–histogram procedure separates partition selection from Bellman iteration. It first
 600 performs one isotonic regression of the initial Bellman target as a function of $\hat{v}(S)$, extracts the flat
 601 regions of the fitted isotonic function as bins, and then runs histogram-based Bellman calibration on
 602 this fixed partition. Thus the method retains the adaptivity of isotonic calibration while restoring the
 603 fixed aggregated-operator structure used in the histogram analysis.

604 Once the partition is fixed, the guarantees of Section 3.2 apply directly, provided the number of bins
 605 produced by the initial isotonic step admits a deterministic high-probability bound. More generally,
 606 the same argument applies to other data-adaptive binning methods that produce a fixed partition
 607 before the Bellman iterations begin.

608 **Theorem 6** (Calibration and Estimation Error for Alg. 2). *Let V_K denote the output of Algorithm 2.*
 609 *Suppose the isotonic regression step produces at most $\hat{B}$ bins, where $\hat{B} \leq B_n$ for some deterministic*
 610 *sequence B_n with probability at least $1 - \delta$. Assume C1–C4 and C5 with $B = B_n$. Then, with*
 611 *probability at least $1 - 2\delta$, the guarantees in Theorems 3 and 4 hold for V_K with $B := B_n$.*

Existing isotonic regression theory suggests that the effective number of induced bins satisfies $\hat{B} = O(n^{1/3})$ when the calibration function is continuously differentiable [16]; this rate can also be enforced by directly constraining the number of constant segments [58]. For the approximation term in Theorem 4 to remain comparable to the initial error, the induced histogram class must also approximate the identity map on the range of $\hat{v}$. This is plausible when $\hat{v}$ is already close to v_0 ; when $\hat{v}$ is substantially miscalibrated, isotonic regression may instead pool poorly predicted regions into a common calibrated value.

D.2 Alternative Calibration Targets

The Bellman calibration criterion studied in the main text calibrates a candidate value predictor v with respect to its own Bellman target,

$$E_\pi[R + \gamma v(S') \mid v(S)].$$

A stronger alternative would instead calibrate v directly against the true value function v_0 . Specifically, we say that v is *strongly Bellman calibrated* if

$$v(S) = E[v_0(S) \mid v(S)] \quad \text{almost surely.}$$

Equivalently, among states assigned the same predicted value, the prediction agrees on average with the true long-run return under π .

This target is conceptually appealing, but substantially less practical. It requires estimating the conditional mean of the unknown quantity $v_0(S)$ given the predicted value $v(S)$. In general, this conditional mean is not directly available from one-step transitions alone and is typically no easier to estimate than the underlying value function itself [28, 29]. We therefore focus on the weaker but directly attainable Bellman-target calibration criterion developed in the main text.

E Experimental results and info

E.1 Synthetic CRM environment

We evaluate all methods in a synthetic customer-relationship-management (CRM) Markov decision process (MDP) that mimics monthly retention and revenue dynamics for a subscription service. The state $s_t \in \mathbb{R}^6$ encodes tenure in months, engagement, fatigue, value segment, price sensitivity, and an indicator of whether the customer is active. At each time t , the agent selects one of three actions: no promotion, light promotion, or strong promotion. Rewards correspond to monthly revenue after discounts and are zero after churn.

Transitions capture key CRM effects. Churn probability depends on tenure, engagement, fatigue, and action via a logistic model; visit probability depends on engagement, fatigue, and action; and revenue conditional on visit scales a baseline value segment by an action-dependent uplift and a price-sensitivity effect, with log-normal noise. Engagement decays over time but is boosted by successful promotions, while fatigue increases with promotion intensity and decays slowly otherwise. Once a customer churns or reaches a maximum tenure of 60 months, the process enters an absorbing state.

We simulate offline datasets of $n_{\text{cust}} = 50,000$ customers over a horizon of $T = 24$ months with discount factor $\gamma = 0.99$. Behavior data are generated by a fixed heuristic policy that sends light promotions by default, suppresses promotions for highly engaged and fatigued customers, and occasionally sends strong promotions to low-engagement, high-value customers. We evaluate off-policy value estimation for a deterministic *aggressive* revenue-seeking target policy that sends strong promotions to low-engagement, high-sensitivity customers and avoids promotions for highly engaged, high-value customers.

Ground-truth values $V^\pi(s_0)$ are estimated via Monte Carlo rollouts in the same environment, and we report $(1 - \gamma)$ -scaled RMSE between the estimated and Monte Carlo values at initial states, averaged across 50 independent runs.

Base estimators. We consider three base estimators: boosted FQE, linear FVI, and neural FVI. The linear model uses a hand-crafted semi-structured feature map consisting of the six raw state variables,

normalized and log-transformed tenure, squared terms for engagement, fatigue, value segment, and price sensitivity, three state-only interaction terms, and additive action one-hot indicators, with no state-action interactions. The neural model takes the raw state concatenated with the action as input and uses a feedforward ReLU network with three hidden layers of width 256. Both FVI estimators are trained by iterating one-step Bellman regressions: the linear model uses ridge-regularized least squares, while the neural model is trained for 100 outer iterations using Adam with learning rate 0.005, one epoch per iteration, and mini-batch size 2000. We do not tune the number of neural iterations, since part of the goal is to illustrate a realistic setting in which a base estimator remains poorly calibrated. Selected neural iterates are saved for the snapshot comparisons. Both FVI estimators use an internal 80/20 train-validation split. The boosted baseline is implemented as a LightGBM FQE model trained on the concatenated state-action input. Under sample splitting, base estimators are fit on the training subset and Bellman calibrators on the held-out calibration subset; we also retrain each base estimator on the full dataset to obtain the *Full Raw* baselines.

For the calibration methods, we use 1000 fitted-value-iteration updates.

E.1.1 Neural snapshots under sample splitting

Tables 3 and 4 report neural FVI snapshot performance under sample splitting for the additional training proportions $p_{\text{train}} = 0.5$ and $p_{\text{train}} = 0.75$. Together, they show how the effect of calibration evolves over training and how it depends on the amount of data reserved for fitting versus calibration.

Iter	Full Raw	Iso	Histogram	Hybrid Iso
10	1.309±0.06	0.581±0.05	0.582±0.05	0.581±0.05
25	0.491±0.13	0.456±0.06	0.456±0.06	0.456±0.06
50	0.320±0.07	0.381±0.05	0.382±0.05	0.382±0.05
100	0.316±0.05	0.350±0.05	0.351±0.05	0.350±0.05

Table 3: Neural snapshot performance with sample splitting for $n = 50,000$ and training proportion $p_{\text{train}} = 0.5$. Lower is better.

Iteration	Full Raw	Iso	Histogram	Hybrid Iso
10	1.28±0.05	0.52±0.05	0.52±0.05	0.52±0.05
25	0.44±0.12	0.38±0.03	0.38±0.04	0.38±0.04
50	0.34±0.06	0.33±0.04	0.33±0.04	0.33±0.04
100	0.33±0.07	0.31±0.04	0.31±0.04	0.31±0.04

Table 4: Neural FVI snapshot results for $n = 50,000$ with $p_{\text{train}} = 0.75$. Lower is better.

E.1.2 Comparing to the uncalibrated (raw) model without retraining on the full data

Table 5 compares calibrated estimators to the corresponding raw model fit only on the training split, so any improvements reflect calibration rather than retraining on the full dataset. This comparison is relevant when retraining on all available data is costly or when a gold-standard calibration dataset is available, such as recent trajectories or on-policy data, alongside a larger historical dataset used for initial fitting. Table 6 further examines this comparison for the neural method across training iterations, illustrating how the effect of calibration evolves over the course of fitting.

F Stationary Measures and Contraction Results

Lemma 1 (Bellman contraction under a stationary measure). *Let P be a Markov operator with stationary distribution μ (i.e., $\mu = \mu P$), and define the Bellman operator $\mathcal{T}f := r + \gamma P f$. Then P is nonexpansive in $L^2(\mu)$:*

$$\|P(f - g)\|_{2,\mu} \leq \|f - g\|_{2,\mu}.$$

Consequently, $\mathcal{T}$ is a γ -contraction:

$$\|\mathcal{T}(f - g)\|_{2,\mu} \leq \gamma \|f - g\|_{2,\mu}.$$

n	Model	Raw	Iso	Hybrid Iso
10,000	Boost	0.681±0.1	0.671±0.1	0.697±0.2
	Lin	0.640±0.04	0.612±0.05	0.641±0.05
	Neur	0.582±0.2	0.550±0.2	0.520±0.1
50,000	Boost	0.433±0.06	0.429±0.06	0.438±0.08
	Lin	0.625±0.02	0.614±0.03	0.616±0.03
	Neur	0.419±0.1	0.383±0.08	0.374±0.08
100,000	Boost	0.360±0.05	0.358±0.05	0.353±0.07
	Lin	0.616±0.02	0.605±0.02	0.606±0.02
	Neur	0.379±0.09	0.351±0.05	0.342±0.05

Table 5: Main results with sample splitting at training proportion $p_{\text{train}} = 0.5$. The raw estimator is fit only on the training split to isolate the effect of calibration.

Iter	Raw	Iso	Quantile	Hybrid Iso
10	1.424±0.1	1.330±0.1	0.735±0.2	0.737±0.2
25	0.665±0.1	0.646±0.1	0.577±0.1	0.571±0.1
50	0.612±0.2	0.585±0.2	0.555±0.1	0.547±0.1
100	0.582±0.2	0.550±0.2	0.524±0.1	0.520±0.1

Table 6: Neural FVI snapshot performance with sample splitting at training proportion $p_{\text{train}} = 0.5$. The raw estimator is fit using only the training split, so differences reflect the effect of calibration rather than access to additional data.

688 *Proof.* Because μ is stationary for P , the operator P is nonexpansive in $L^2(\mu)$. Jensen’s inequality
689 implies

$$(Ph)^2 \leq P(h^2) \quad \text{pointwise.}$$

690 Integrating both sides with respect to μ and using $\mu = \mu P$ yields

$$\|Ph\|_{2,\mu}^2 = \int (Ph)^2 d\mu \leq \int P(h^2) d\mu = \int h^2 d\mu = \|h\|_{2,\mu}^2.$$

691 Now apply this with $h = f - g$. Since the reward function r cancels,

$$\mathcal{T}f - \mathcal{T}g = \gamma P(f - g).$$

692 Therefore,

$$\|\mathcal{T}f - \mathcal{T}g\|_{2,\mu} = \gamma \|P(f - g)\|_{2,\mu} \leq \gamma \|f - g\|_{2,\mu},$$

693 which proves the claim. $\square$

694 **Lemma 2** (Stationarity under coarsening). *Let P be a Markov kernel on $\mathcal{S}$ and let λ be stationary*
695 *for P (i.e., $\lambda P = \lambda$). For any measurable coarsening map $g : \mathcal{S} \rightarrow \mathbb{R}$, define*

$$P_g f(s) := \mathbb{E}[f(S') \mid g(S) = g(s)],$$

696 *where (S, S') has joint law $\lambda(ds) P(s, ds')$. Then λ is stationary for P_g .*

697 *Proof.* Let $(S, S') \sim \lambda(ds) P(s, ds')$, so $S \sim \lambda$ and $S' \mid S \sim P(S, \cdot)$. Then

$$\begin{aligned} \int P_g f(s) \lambda(ds) &= \mathbb{E}_{\lambda, P}[P_g f(S)] \\ &= \mathbb{E}_{\lambda, P}[\mathbb{E}_{\lambda, P}[f(S') \mid g(S)]] \\ &= \mathbb{E}_{\lambda, P}[f(S')]. \end{aligned}$$

698 Since λ is stationary for P , the marginal of S' is again λ , hence $\mathbb{E}_{\lambda, P}[f(S')] = \int f(s) \lambda(ds)$. $\square$

699 G Calibration–Refinement Proofs

700 In this section, set

$$\mathcal{B}_{\pi, \hat{v}} q := \Pi_{\hat{v}} \mathcal{T}_{\pi} q, \quad V_{\hat{v}}^* \in \mathcal{V}_{\hat{v}}, \quad V_{\hat{v}}^* = \mathcal{B}_{\pi, \hat{v}} V_{\hat{v}}^*.$$

701 **Theorem 7** (Sharper calibration–refinement bound). *Under A1,*

$$\|\hat{v} - v_0\|_{2, \rho} \leq \|\hat{v} - V_{\hat{v}}^*\|_{2, \rho} + \frac{\sqrt{\mathcal{C}_{\hat{v}, \gamma}}}{1 - \gamma} \min_{\theta: \mathbb{R} \rightarrow \mathbb{R}} \|\theta \circ \hat{v} - v_0\|_{2, \rho}.$$

702 *Proof of Theorem 7.* Define $P_{\pi, \hat{v}} := \Pi_{\hat{v}} P_{\pi}$ and $r_{\pi, \hat{v}} := \Pi_{\hat{v}} r_{\pi}$. Because v_0 and $V_{\hat{v}}^*$ satisfy the
703 fixed-point equations

$$v_0 = \mathcal{T}_{\pi} v_0, \quad V_{\hat{v}}^* = \mathcal{B}_{\pi, \hat{v}} V_{\hat{v}}^* = r_{\pi, \hat{v}} + \gamma P_{\pi, \hat{v}} V_{\hat{v}}^*,$$

704 we begin by writing

$$V_{\hat{v}}^* - v_0 = (\mathcal{B}_{\pi, \hat{v}} V_{\hat{v}}^* - \mathcal{B}_{\pi, \hat{v}} v_0) + (\mathcal{B}_{\pi, \hat{v}} v_0 - \mathcal{T}_{\pi} v_0).$$

705 Using linearity on differences,

$$\mathcal{B}_{\pi, \hat{v}} V_{\hat{v}}^* - \mathcal{B}_{\pi, \hat{v}} v_0 = \gamma P_{\pi, \hat{v}} (V_{\hat{v}}^* - v_0),$$

706 so

$$V_{\hat{v}}^* - v_0 = \gamma P_{\pi, \hat{v}} (V_{\hat{v}}^* - v_0) + (\mathcal{B}_{\pi, \hat{v}} v_0 - v_0),$$

707 where we used $\mathcal{T}_{\pi} v_0 = v_0$ for the second term. Since $\mathcal{B}_{\pi, \hat{v}} v_0 = \Pi_{\hat{v}} v_0$, this gives

$$V_{\hat{v}}^* - v_0 = \gamma P_{\pi, \hat{v}} (V_{\hat{v}}^* - v_0) + (\Pi_{\hat{v}} - I) v_0.$$

708 Let $a := (\Pi_{\hat{v}} - I) v_0$. Iterating the display yields

$$V_{\hat{v}}^* - v_0 = \sum_{t=0}^{\infty} \gamma^t P_{\pi, \hat{v}}^t a.$$

709 For each $t \geq 0$, set

$$c_t := \left\| \frac{d(\rho P_{\pi, \hat{v}}^t)}{d\rho} \right\|_{\infty}.$$

710 Jensen’s inequality implies

$$\|P_{\pi, \hat{v}}^t a\|_{2, \rho} \leq \sqrt{c_t} \|a\|_{2, \rho}.$$

711 Therefore,

$$\|V_{\hat{v}}^* - v_0\|_{2, \rho} \leq \frac{\sqrt{\mathcal{C}_{\hat{v}, \gamma}}}{1 - \gamma} \|\Pi_{\hat{v}} v_0 - v_0\|_{2, \rho}. \quad (\star)$$

712 Since $\Pi_{\hat{v}}$ is the $L^2(\rho)$ -projection onto $\{\theta \circ \hat{v} : \theta : \mathbb{R} \rightarrow \mathbb{R}\}$,

$$\|\Pi_{\hat{v}} v_0 - v_0\|_{2, \rho} = \min_{\theta: \mathbb{R} \rightarrow \mathbb{R}} \|\theta \circ \hat{v} - v_0\|_{2, \rho}.$$

713 Finally, by the triangle inequality,

$$\|\hat{v} - v_0\|_{2, \rho} \leq \|\hat{v} - V_{\hat{v}}^*\|_{2, \rho} + \|V_{\hat{v}}^* - v_0\|_{2, \rho},$$

714 Combining this display with $(\star)$ yields the stated result. $\square$

715 *Proof of Theorem 1.* Let $P_{\pi, \hat{v}} := \Pi_{\hat{v}} P_{\pi}$. As in the proof of Theorem 7,

$$\|V_{\hat{v}}^* - v_0\|_{2, \rho} \leq \frac{\sqrt{\mathcal{C}_{\hat{v}, \gamma}}}{1 - \gamma} \min_{\theta: \mathbb{R} \rightarrow \mathbb{R}} \|\theta \circ \hat{v} - v_0\|_{2, \rho}. \quad (\star)$$

716 Moreover,

$$\hat{v} - V_{\hat{v}}^* = \hat{v} - \mathcal{B}_{\pi, \hat{v}} \hat{v} + \gamma P_{\pi, \hat{v}} (\hat{v} - V_{\hat{v}}^*),$$

717 and therefore

$$\hat{v} - V_{\hat{v}}^* = \sum_{t=0}^{\infty} \gamma^t P_{\pi, \hat{v}}^t \{\hat{v} - \mathcal{B}_{\pi, \hat{v}} \hat{v}\}.$$

718 Since $\mathcal{B}_{\pi, \hat{v}} \hat{v} = \Gamma_0(\hat{v})$, the concentrability argument above yields

$$\|\hat{v} - V_{\hat{v}}^*\|_{2, \rho} \leq \frac{\sqrt{\mathcal{C}_{\hat{v}, \gamma}}}{1 - \gamma} \text{Cal}_{\ell^2}(\hat{v}). \quad (\dagger)$$

719 The triangle inequality and displays $(\star)$ and $(\dagger)$ give the claim. $\square$

720 H Proof of Theorem 2

721 *Proof of Theorem 2.* Fix v and treat the nuisance estimates as fixed functions. By definition,

$$\widehat{\mathcal{T}}_\pi v(s) = \mathbb{E} \left[(\pi \hat{q}_v)(S) + \widehat{w}_\pi(A | S) \{R + \gamma v(S') - \hat{q}_v(S, A)\} \mid S = s \right].$$

722 Since $(\pi \hat{q}_v)(S)$ depends only on S ,

$$\mathbb{E} [(\pi \hat{q}_v)(S) \mid S = s] = (\pi \hat{q}_v)(s).$$

723 For the second term, using the definition of b_0 and $q_v := r_0 + \gamma P v$,

$$\begin{aligned} & \mathbb{E} \left[\widehat{w}_\pi(A | S) \{R + \gamma v(S') - \hat{q}_v(S, A)\} \mid S = s \right] \\ &= b_0 \left\{ \widehat{w}_\pi(\cdot | s) \mathbb{E} [R + \gamma v(S') - \hat{q}_v(s, \cdot) \mid S = s, A = \cdot] \right\}(s) \\ &= b_0 \{ \widehat{w}_\pi(q_v - \hat{q}_v) \}(s). \end{aligned}$$

724 Hence

$$\widehat{\mathcal{T}}_\pi v(s) = (\pi \hat{q}_v)(s) + b_0 \{ \widehat{w}_\pi(q_v - \hat{q}_v) \}(s).$$

725 Subtracting $(\mathcal{T}_\pi v)(s) = (\pi q_v)(s)$ yields

$$\begin{aligned} \widehat{\mathcal{T}}_\pi v(s) - (\mathcal{T}_\pi v)(s) &= (\pi \hat{q}_v - \pi q_v)(s) + b_0 \{ \widehat{w}_\pi(q_v - \hat{q}_v) \}(s) \\ &= (\pi(\hat{q}_v - q_v))(s) - b_0 \{ \widehat{w}_\pi(\hat{q}_v - q_v) \}(s). \end{aligned}$$

726 Using $(\pi f)(s) = b_0 \{ w_\pi f \}(s)$ with $f = \hat{q}_v - q_v$ gives

$$(\pi(\hat{q}_v - q_v))(s) = b_0 \{ w_\pi(\hat{q}_v - q_v) \}(s),$$

727 so

$$\widehat{\mathcal{T}}_\pi v(s) - (\mathcal{T}_\pi v)(s) = b_0 \{ (w_\pi - \widehat{w}_\pi)(\hat{q}_v - q_v) \}(s).$$

728 Taking conditional expectations given $v(S) = t$ proves the claim. $\square$

729 I Notation and Maximal Inequalities

730 Let P_0 denote the joint distribution of (S, A, R, S') induced by the behavior policy, and let P_n denote
731 the empirical measure of the calibration sample $\mathcal{C}_n$.

732 We define empirical L^2 norms with respect to the state and next-state samples. For any state function
733 f , let

$$\|f\|_{n,S} := \left(\frac{1}{n} \sum_{i=1}^n f(S_i)^2 \right)^{1/2}, \quad \|f\|_{n,S'} := \left(\frac{1}{n} \sum_{i=1}^n f(S'_i)^2 \right)^{1/2},$$

734 where S_i and S'_i denote the observed states and next states, respectively, in the calibration sample $\mathcal{C}_n$.

735 Further define

$$\widehat{\mathcal{G}} := \left\{ (f_1 - f_2)(\widehat{Y}_{f_2} - f_2) : f_1, f_2 \in \mathcal{H}_B(\hat{v}) \right\}.$$

736 By assumption, both $\hat{v}$ and the Bellman target-estimate map $v \mapsto \widehat{Y}_v$ are fixed conditional on the
737 training data, which is independent of the calibration sample $\mathcal{C}_n$. Consequently, the classes $\mathcal{H}_B(\hat{v})$
738 and $\widehat{\mathcal{G}}$ are non-random conditional on the training dataset.

739 For any distribution Q and any uniformly bounded function class $\mathcal{F}$, let $N(\varepsilon, \mathcal{F}, L^2(Q))$ denote the
740 ε -covering number of $\mathcal{F}$ under the $L^2(Q)$ norm [64]. Define the uniform entropy integral of $\mathcal{F}$ by

$$\mathcal{J}(\delta, \mathcal{F}) := \int_0^\delta \sup_Q \sqrt{\log N(\epsilon, \mathcal{F}, L^2(Q))} d\epsilon,$$

741 where the supremum is taken over all discrete probability distributions Q .

742 Finally, for two quantities x and y , we write $x \lesssim y$ to mean that x is bounded above by y up to a
743 universal constant that depends only on global constants appearing in our conditions.

744 **I.1 Local maximal inequality**

745 Let $O_1, \dots, O_n \in \mathcal{O}$ be independent random variables. For any function $f : \mathcal{O} \rightarrow \mathbb{R}$, define

$$\|f\|_{2,\bar{P}} := \sqrt{\frac{1}{n} \sum_{i=1}^n \mathbb{E}[f(O_i)^2]}. \quad (4)$$

746 For a star-shaped class of functions $\mathcal{F}$ and a radius $\delta \in (0, \infty)$, define the localized Rademacher
747 complexity

$$\mathcal{R}_n(\mathcal{F}, \delta) := \mathbb{E} \left[\sup_{\substack{f \in \mathcal{F} \\ \|f\|_{2,\bar{P}} \leq \delta}} \frac{1}{n} \sum_{i=1}^n \epsilon_i f(O_i) \right],$$

748 where ϵ_i are i.i.d. Rademacher random variables.

749 The following lemma provides a local maximal inequality and restates Lemma 11 of [19].

750 **Lemma 3** (Local maximal inequality). *Let $\mathcal{F}$ be a star-shaped class of functions satisfying*
751 *$\sup_{f \in \mathcal{F}} \|f\|_\infty \leq M$. Let $\delta > 0$ satisfy the critical radius condition $\mathcal{R}_n(\mathcal{F}, \delta) \leq \delta^2$. Suppose*
752 *further that $n^{-1/2} \sqrt{\log \log(1/\delta)} = o(\delta)$. Then there exists a universal constant $C > 0$ such that,*
753 *for all $u \geq 1$, with probability at least $1 - e^{-u^2}$, every $f \in \mathcal{F}$ satisfies*

$$\frac{1}{n} \sum_{i=1}^n (f(O_i) - \mathbb{E}[f(O_i)]) \leq C \left(\delta^2 + \delta \|f\|_{2,\bar{P}} + \frac{u \|f\|_{2,\bar{P}}}{\sqrt{n}} + \frac{Mu^2}{n} \right).$$

754 *Proof.* This is Lemma 11 of Foster and Syrgkanis [19], specialized to the notation above. $\square$

755 The following lemma bounds the localized Rademacher complexity in terms of the uniform entropy
756 integral and is a direct consequence of Theorem 2.1 of Van Der Vaart and Wellner [63].

757 **Lemma 4.** *Let $\mathcal{F}$ be a star-shaped class of functions such that $\sup_{f \in \mathcal{F}} \|f\|_\infty \leq M$. Then, for every*
758 *$\delta > 0$,*

$$\mathcal{R}_n(\mathcal{F}, \delta) \lesssim \frac{1}{\sqrt{n}} \mathcal{J}(\delta, \mathcal{F}) \left(1 + \frac{\mathcal{J}(\delta, \mathcal{F})}{\delta \sqrt{n}} \right),$$

759 *where the implicit constant depends only on M .*

760 *Proof.* This bound follows directly from the argument in the proof of Theorem 2.1 of Van Der Vaart
761 and Wellner [63]; see in particular the step where the local Rademacher complexity is controlled by
762 the uniform entropy integral for star-shaped classes. $\square$

763 **J Proof of Theorem 3**

764 **J.1 Technical lemmas**

765 **Lemma 5.** *Under our conditions,*

$$\mathcal{J}(\delta, \hat{\mathcal{G}}) \lesssim \delta \sqrt{B \log(1/\delta)},$$

766 *where the implicit constant is independent of B .*

767 *Proof.* By Condition C3,

$$\|\hat{q}_f - \hat{q}_g\|_{n,S,A} \leq L \|f - g\|_{n,S'}$$

768 almost surely. For each s , by Jensen's inequality,

$$|\pi h(s)|^2 = \left| \sum_a \pi(a | s) h(a, s) \right|^2 \leq \sum_a \pi(a | s) |h(a, s)|^2,$$

769 so $\|\pi h\|_{n,S} \leq \|h\|_{n,S,A}$. Hence, we also have

$$\|\pi(\hat{q}_f - \hat{q}_g)\|_{n,S} \leq \|\hat{q}_f - \hat{q}_g\|_{n,S,A} \leq L\|f - g\|_{n,S'}.$$

770 Hence, by the definition of $\hat{Y}_f$ and boundedness of $\hat{w}_\pi$,

$$\|\hat{Y}_f - \hat{Y}_g\|_{L^2(P_n)} \lesssim \|f - g\|_{n,S'}.$$

771 Take $f, g \in \hat{\mathcal{G}}$ with $f = (f_1 - f_2)(\hat{Y}_{f_2} - f_2)$ and $g = (g_1 - g_2)(\hat{Y}_{g_2} - g_2)$ for $f_1, f_2, g_1, g_2 \in \mathcal{H}_B(\hat{v})$.
 772 By Lipschitz continuity of multiplication and boundedness of nuisances, we have

$$\|f - g\|_{L^2(P_n)} \lesssim \|f_1 - g_1\|_{n,S} + \|f_2 - g_2\|_{n,S} + \|f_2 - g_2\|_{n,S'}.$$

773 Taking the supremum over all discrete distributions Q , we obtain

$$\|f - g\|_{L^2(P_n)} \lesssim \sup_Q \|f_1 - g_1\|_{L^2(Q)} + \sup_Q \|f_2 - g_2\|_{L^2(Q)}.$$

774 Hence, by preservation of entropy integrals in Van Der Vaart and Wellner [64],

$$\log N(\varepsilon, \hat{\mathcal{G}}, L^2(P_n)) \lesssim \sup_Q \log N(\varepsilon, \mathcal{H}_B(\hat{v}), L^2(Q)).$$

775 Taking the supremum over discrete distributions Q on both sides yields the uniform covering number
 776 bound

$$\sup_Q \log N(\varepsilon, \hat{\mathcal{G}}, L^2(Q)) \lesssim \sup_Q \log N(\varepsilon, \mathcal{H}_B(\hat{v}), L^2(Q)).$$

777 The class $\mathcal{H}_B(\hat{v})$ satisfies

$$\sup_Q \log N(\varepsilon, \mathcal{H}_B(\hat{v}), L^2(Q)) \leq \sup_Q \log N(\varepsilon, \tilde{\mathcal{F}}_B, L^2(Q \circ \hat{v}^{-1})),$$

778 where $Q \circ \hat{v}^{-1}$ denotes the pushforward of Q under $\hat{v}$. Hence,

$$\sup_Q \log N(\varepsilon, \mathcal{H}_B(\hat{v}), L^2(Q)) \leq \sup_Q \log N(\varepsilon, \tilde{\mathcal{F}}_B, L^2(Q)),$$

779 where, by a slight abuse of notation, the supremum on the right-hand side is taken over all discrete
 780 probability distributions Q on $\mathbb{R}$. The class $\tilde{\mathcal{F}}_B$ consists of all piecewise-constant functions on $\mathbb{R}$
 781 taking at most B values. Therefore, $\tilde{\mathcal{F}}_B$ has VC-subgraph dimension $O(B)$, and Van Der Vaart and
 782 Wellner [64] implies

$$\begin{aligned} \sup_Q \log N(\varepsilon, \mathcal{H}_B(\hat{v}), L^2(Q)) &\lesssim \sup_Q \log N(\varepsilon, \tilde{\mathcal{F}}_B, L^2(Q)) \\ &\lesssim B \log(1/\varepsilon). \end{aligned}$$

783 Consequently, $\mathcal{J}(\delta, \mathcal{H}_B(\hat{v})) \lesssim \delta \sqrt{B \log(1/\delta)}$. □

784 J.2 Proof of Theorem 3

785 *Proof of Theorem 3.* Denote $O_i := (S_i, A_i, R_i, S'_i)$. The first-order optimality conditions for $\theta_n^{(K)}$
 786 imply that, for each bin $b \in [B]$,

$$\frac{1}{n} \sum_{i=1}^n \mathbf{1}_{I_b}(\hat{v}(S_i)) \{\hat{Y}_{V_{K-1}}(O_i) - V_K(S_i)\} = 0.$$

787 Hence, for any $f \in \mathcal{G}_B$, which is a linear combination of these indicators,

$$\frac{1}{n} \sum_{i=1}^n f(\hat{v}(S_i)) \{\hat{Y}_{V_{K-1}}(O_i) - V_K(S_i)\} = 0. \quad (5)$$

788 Moreover, for any function $g : \mathbb{R} \rightarrow \mathbb{R}$, we may write $g \circ V_K = f \circ \hat{v}$ for some f , and therefore, for
 789 all such g ,

$$\frac{1}{n} \sum_{i=1}^n g(V_K(S_i)) \{\hat{Y}_{V_{K-1}}(O_i) - V_K(S_i)\} = 0.$$

790 Noting that $\Gamma_0(V_K) - V_K$ is of the form $g \circ V_K$ for some function g , we have

$$0 = \frac{1}{n} \sum_{i=1}^n \left\{ \Gamma_0(V_K)(S_i) - V_K(S_i) \right\} \\ \times \left\{ \hat{Y}_{V_{K-1}}(O_i) - V_K(S_i) \right\}.$$

791 Adding and subtracting P_0 yields

$$P_0 \left\{ \Gamma_0(V_K) - V_K \right\} \left\{ \hat{Y}_{V_K} - V_K \right\} \\ = (P_0 - P_n) \left\{ \Gamma_0(V_K) - V_K \right\} \left\{ \hat{Y}_{V_{K-1}} - V_K \right\} \\ + P_0 \left\{ \Gamma_0(V_K) - V_K \right\} \left\{ \hat{Y}_{V_K} - \hat{Y}_{V_{K-1}} \right\}. \quad (6)$$

792 We rewrite the left-hand side of (6). Adding and subtracting and applying the law of total expectation,
793 the calibration error decomposes as

$$\| \Gamma_0(V_K) - V_K \|^2 \\ = P_0 \left\{ \Gamma_0(V_K) - V_K \right\} \left\{ \hat{Y}_{V_K} - V_K \right\} \\ + P_0 \left\{ \Gamma_0(V_K) - V_K \right\} \left\{ \Gamma_0(V_K) - \hat{Y}_{V_K} \right\}. \quad (7)$$

794 The first term on the right-hand side can be decomposed as in (6). By the law of total expectation, the
795 proof of Theorem 2, and the Cauchy–Schwarz inequality, the second term satisfies

$$\left| P_0 \left\{ \Gamma_0(V_K) - V_K \right\} \left\{ \hat{Y}_{V_K} - \Gamma_0(V_K) \right\} \right| \\ = \left| P_0 \left\{ \Gamma_0(V_K) - V_K \right\} \left\{ \hat{Y}_{V_K} - \mathcal{T}_\pi(V_K) \right\} \right| \\ \leq \| \Gamma_0(V_K) - V_K \| \left\| b_0 \{ (\hat{w}_\pi - w_\pi) (\hat{q}_{V_K} - q_{V_K}) \} \right\|_{2,\rho}.$$

796 Hence, (6) and (7) together imply that

$$\| \Gamma_0(V_K) - V_K \|^2 \\ \leq (P_0 - P_n) \left\{ \Gamma_0(V_K) - V_K \right\} \left\{ \hat{Y}_{V_{K-1}} - V_K \right\} \\ + \| \Gamma_0(V_K) - V_K \| \left\| \hat{\mathcal{T}}_\pi V_K - \hat{\mathcal{T}}_\pi V_{K-1} \right\| \\ + \| \Gamma_0(V_K) - V_K \| \left\| b_0 \{ (\hat{w}_\pi - w_\pi) (\hat{q}_{V_K} - q_{V_K}) \} \right\|_{2,\rho}. \quad (8)$$

797 where $\hat{\mathcal{T}}_\pi v(s) := \mathbb{E}[\hat{Y}_v(O) \mid S = s]$. Here, the second term on the right-hand side follows from (6),
798 noting that

$$\left| P_0 \left\{ \Gamma_0(V_K) - V_K \right\} \left\{ \hat{Y}_{V_K} - \hat{Y}_{V_{K-1}} \right\} \right| \\ = \left| P_0 \left\{ \Gamma_0(V_K) - V_K \right\} \mathbb{E} \left[\hat{Y}_{V_K} - \hat{Y}_{V_{K-1}} \mid S \right] \right| \\ \leq \| \Gamma_0(V_K) - V_K \| \left\| \mathbb{E} \left[\hat{Y}_{V_K} - \hat{Y}_{V_{K-1}} \mid S \right] \right\| \\ = \| \Gamma_0(V_K) - V_K \| \left\| \hat{\mathcal{T}}_\pi V_K - \hat{\mathcal{T}}_\pi V_{K-1} \right\|_{2,\rho}.$$

799 We now turn to bounding the empirical process term on the right-hand side of (8). Observe that

$$(\Gamma_0(V_K) - V_K)(\hat{Y}_{V_{K-1}} - V_K)$$

800 lies in a uniformly bounded subset of the class

$$\widehat{\mathcal{G}} := \{(f_1 - f_2)(\widehat{Y}_{f_2} - f_2) : f_1, f_2 \in \mathcal{H}_B(\widehat{v})\},$$

801 By Lemma 5, it holds that $\mathcal{J}(\delta, \widehat{\mathcal{G}}) \lesssim \delta \sqrt{B \log(1/\delta)}$. By assumption, $\widehat{\mathcal{G}}$ is a fixed, nonrandom
 802 function class conditional on the training data, which is independent of the calibration sample $\mathcal{C}_n$.
 803 Hence, applying Lemma 4 conditional on the training data, we obtain

$$\begin{aligned} \mathcal{R}_n(\widehat{\mathcal{G}}, \delta) &\lesssim \frac{1}{\sqrt{n}} \mathcal{J}(\delta, \widehat{\mathcal{G}}) \left(1 + \frac{\mathcal{J}(\delta, \widehat{\mathcal{G}})}{\delta \sqrt{n}}\right) \\ &\lesssim \frac{1}{\sqrt{n}} \delta \sqrt{B \log(1/\delta)} \left(1 + \frac{\delta \sqrt{B \log(1/\delta)}}{\delta \sqrt{n}}\right) \\ &\lesssim \frac{\delta \sqrt{B \log(1/\delta)}}{\sqrt{n}} + \frac{B \log(1/\delta)}{n}. \end{aligned}$$

804 The critical radius

$$\delta_n := \sqrt{\frac{B}{n} \log\left(\frac{n}{B}\right)},$$

805 satisfies

$$\delta_n = \inf\{\delta > 0 : \mathcal{R}_n(\widehat{\mathcal{G}}, \delta) \lesssim \delta^2\}.$$

806 Applying Lemma 3 conditional on the training data with $\mathcal{F} := \widehat{\mathcal{G}}$, we conclude that the following
 807 holds with probability at least $1 - e^{-u^2}$ for every $f \in \widehat{\mathcal{G}}$:

$$\frac{1}{n} \sum_{i=1}^n (f(O_i) - \mathbb{E}[f(O_i)]) \lesssim \delta_n^2 + \delta_n \|f\|_{2, P_0} + \frac{u \|f\|_{2, P_0}}{\sqrt{n}} + \frac{u^2}{n}.$$

808 Choosing $u = \sqrt{\log(1/\eta)}$ gives $1 - e^{-u^2} = 1 - \eta$. Hence, with probability at least $1 - \eta$,

$$\begin{aligned} &\frac{1}{n} \sum_{i=1}^n (f(O_i) - \mathbb{E}[f(O_i)]) \\ &\lesssim \delta_n^2 + \delta_n \|f\|_{2, P_0} + \frac{\sqrt{\log(1/\eta)} \|f\|_{2, P_0}}{\sqrt{n}} + \frac{\log(1/\eta)}{n}. \end{aligned}$$

809 By boundedness of our nuisances,

$$\|\{\Gamma_0(V_K) - V_K\} \{\widehat{Y}_{V_{K-1}} - V_K\}\| \lesssim \|\Gamma_0(V_K) - V_K\|_{2, \rho}.$$

810 Hence, with probability at least $1 - \eta$,

$$\begin{aligned} &(P_n - P_0) \left\{ \Gamma_0(V_K) - V_K \right\} \left\{ \widehat{Y}_{V_{K-1}} - V_K \right\} \\ &\lesssim \delta_n^2 + \delta_n \|\Gamma_0(V_K) - V_K\| \\ &\quad + \frac{\sqrt{\log(1/\eta)} \|\Gamma_0(V_K) - V_K\|}{\sqrt{n}} + \frac{\log(1/\eta)}{n}, \end{aligned}$$

811 where the implicit constants do not depend on B .

812 Combining the above with (8), we find with probability at least $1 - \eta$,

$$\begin{aligned} &\|\Gamma_0(V_K) - V_K\|^2 \\ &\lesssim \delta_n^2 + \delta_n \|\Gamma_0(V_K) - V_K\| \\ &\quad + \frac{\sqrt{\log(1/\eta)} \|\Gamma_0(V_K) - V_K\|}{\sqrt{n}} + \frac{\log(1/\eta)}{n} \\ &\quad + \|\Gamma_0(V_K) - V_K\| \|\widehat{\mathcal{T}}_\pi V_K - \widehat{\mathcal{T}}_\pi V_{K-1}\| \\ &\quad + \|\Gamma_0(V_K) - V_K\| \|b_0\{(\widehat{w}_\pi - w_\pi)(\widehat{q}_{V_K} - q_{V_K})\}\|_{2, \rho}. \end{aligned} \tag{9}$$

813 The inequality in (9) implies that, with probability at least $1 - \eta$, the calibration error satisfies

$$\begin{aligned} \|\Gamma_0(V_K) - V_K\| &\lesssim \delta_n + \sqrt{\frac{\log(1/\eta)}{n}} \\ &\quad + \|\widehat{\mathcal{T}}_\pi V_K - \widehat{\mathcal{T}}_\pi V_{K-1}\| \\ &\quad + \|b_0\{(\widehat{w}_\pi - w_\pi)(\widehat{q}_{V_K} - q_{V_K})\}\|_{2,\rho}. \end{aligned}$$

814 Recall that $\delta_n := \sqrt{\frac{B}{n} \log\left(\frac{n}{B}\right)}$. Then, with probability at least $1 - \eta$,

$$\begin{aligned} \|\Gamma_0(V_K) - V_K\| &\lesssim \sqrt{\frac{B}{n} \log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(1/\eta)}{n}} \\ &\quad + \|\widehat{\mathcal{T}}_\pi V_K - \widehat{\mathcal{T}}_\pi V_{K-1}\| \\ &\quad + \|b_0\{(\widehat{w}_\pi - w_\pi)(\widehat{q}_{V_K} - q_{V_K})\}\|_{2,\rho}. \end{aligned}$$

815

□

816 **K Proof of Theorem 4**

817 **K.1 Additional notation**

818 Define the bin-index map $\hat{b}(s)$ by setting $\hat{b}(s) = b$ whenever $\hat{v}(s) \in I_b$. Let Π_B denote the binning
819 projection

$$(\Pi_B g)(s) := E\left[g(S) \mid \hat{v}(S) \in I_{\hat{b}(s)}\right].$$

820 Denote

$$\mathcal{B}_{\pi,B}(g)(s) := (\Pi_B \mathcal{T}_\pi g)(s).$$

821 Define V_B^* as the fixed point satisfying

$$\mathcal{B}_{\pi,B}(V_B^*) = V_B^*.$$

822 Such a fixed point exists by Banach's fixed point theorem because $\mathcal{B}_{\pi,B}(g) \in \mathcal{H}_B(\hat{v})$ whenever
823 $g \in \mathcal{H}_B(\hat{v})$, and the operator is a γ -contraction in the sup-norm.

824 Recall the projected transition operator

$$(P_{\pi,B} h)(s) = \mathbb{E}_\pi \left[h(S') \mid \hat{v}(S) \in I_{\hat{b}(s)} \right].$$

825 It holds that

$$P_{\pi,B} := \Pi_B P_\pi,$$

826 and

$$\Pi_B \mathcal{T}_\pi(f) - \Pi_B \mathcal{T}_\pi(g) = \gamma P_{\pi,B}(f - g).$$

827 **K.2 Supporting lemmas**

828 We begin with the deterministic ingredients for the histogram value bound.

829 Throughout, histogram bins mean the retained cells of the fitted partition: empty empirical cells are
830 merged with a neighboring cell or discarded, and B denotes the number of retained cells. Population
831 projections are defined on the corresponding non-null cells; ρ -null cells are identified in $L^2(\rho)$ and
832 omitted from the index set.

833 **Lemma 6** (Sup-norm contraction of the aggregated Bellman operator). *For any bounded $f, g : \mathcal{S} \rightarrow$
834 $\mathbb{R}$,*

$$\|\mathcal{B}_{\pi,B} f - \mathcal{B}_{\pi,B} g\|_\infty \leq \gamma \|f - g\|_\infty.$$

835 *Proof.* Since the rewards cancel on differences,

$$\mathcal{B}_{\pi,B}f - \mathcal{B}_{\pi,B}g = \gamma P_{\pi,B}(f - g).$$

836 The operator $P_{\pi,B}$ is a Markov averaging operator. Hence, for any bounded h ,

$$|P_{\pi,B}h(s)| \leq E_{\pi}[|h(S')| \mid \hat{v}(S) \in I_{\hat{b}(s)}] \leq \|h\|_{\infty}.$$

837 Taking $h = f - g$ proves the claim. □

838 **Lemma 7** (Approximation error under aggregated concentrability). *Assume C4. Then*

$$\|V_B^* - v_0\|_{2,\rho} \leq \frac{\sqrt{\mathcal{C}_{B,\gamma}}}{1 - \gamma} \|(I - \Pi_B)v_0\|_{2,\rho}.$$

839 *Proof.* Because $v_0 = \mathcal{T}_{\pi}v_0$ and $V_B^* = \Pi_B \mathcal{T}_{\pi}V_B^*$,

$$V_B^* - v_0 = \gamma P_{\pi,B}(V_B^* - v_0) + (\Pi_B - I)v_0.$$

840 Let $a := (\Pi_B - I)v_0$. Iterating the display and using boundedness gives the resolvent expansion

$$V_B^* - v_0 = \sum_{t=0}^{\infty} \gamma^t P_{\pi,B}^t a.$$

841 For any $t \geq 0$, set

$$c_t := \left\| \frac{d(\rho P_{\pi,B}^t)}{d\rho} \right\|_{\infty}.$$

842 Jensen's inequality for the Markov operator $P_{\pi,B}^t$ gives

$$\|P_{\pi,B}^t a\|_{2,\rho}^2 \leq \int P_{\pi,B}^t(a^2) d\rho = \int a^2 d(\rho P_{\pi,B}^t) \leq c_t \|a\|_{2,\rho}^2.$$

843 Therefore,

$$\|V_B^* - v_0\|_{2,\rho} \leq \sum_{t=0}^{\infty} \gamma^t \sqrt{c_t} \|a\|_{2,\rho} = \frac{\sqrt{\mathcal{C}_{B,\gamma}}}{1 - \gamma} \|(I - \Pi_B)v_0\|_{2,\rho}.$$

844 □

845 **Lemma 8** (Inexact iterations under aggregated concentrability). *Assume C4. Let $\{\eta_k\}_{k \geq 1}$ be any*
 846 *sequence such that*

$$\|V_k - \mathcal{B}_{\pi,B}(V_{k-1})\|_{2,\rho} \leq \eta_k \quad \text{for all } k.$$

847 For $m \geq 0$, set

$$c_m := \left\| \frac{d(\rho P_{\pi,B}^m)}{d\rho} \right\|_{\infty}.$$

848 Then, for any $K \geq 1$,

$$\|V_K - V_B^*\|_{2,\rho} \leq \gamma^K \|\hat{v} - V_B^*\|_{\infty} + \sum_{j=1}^K \gamma^{K-j} \sqrt{c_{K-j}} \eta_j.$$

849 Consequently,

$$\|V_K - V_B^*\|_{2,\rho} \leq \gamma^K \|\hat{v} - V_B^*\|_{\infty} + \frac{\sqrt{\mathcal{C}_{B,\gamma}}}{1 - \gamma} \max_{1 \leq j \leq K} \eta_j.$$

850 *Proof.* Let

$$e_k := V_k - \mathcal{B}_{\pi,B}(V_{k-1}).$$

851 Since $V_B^* = \mathcal{B}_{\pi,B}V_B^*$,

$$V_k - V_B^* = \gamma P_{\pi,B}(V_{k-1} - V_B^*) + e_k.$$

852 Iterating this recursion yields

$$V_K - V_B^* = \gamma^K P_{\pi,B}^K (\hat{v} - V_B^*) + \sum_{j=1}^K \gamma^{K-j} P_{\pi,B}^{K-j} e_j.$$

853 The first term is bounded by $\gamma^K \|\hat{v} - V_B^*\|_\infty$ because $P_{\pi,B}$ is a Markov operator. For the remaining
854 terms, the same Jensen and change-of-measure argument used in Lemma 7 gives

$$\|P_{\pi,B}^{K-j} e_j\|_{2,\rho} \leq \sqrt{c_{K-j}} \|e_j\|_{2,\rho} \leq \sqrt{c_{K-j}} \eta_j.$$

855 The displayed bounds follow by the triangle inequality and the definition of $\mathcal{C}_{B,\gamma}$. $\square$

856 **Lemma 9** (Error bound for inexact updates). *There exists $C < \infty$ such that, with probability at least*
857 *$1 - \delta$, uniformly over $k = 1, \dots, K$,*

$$\begin{aligned} & \|V_k - \mathcal{B}_{\pi,B}(V_{k-1})\|_{2,\rho} \\ & \leq C \left\{ \sqrt{\frac{B}{n} \log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(K/\delta)}{n}} + \|b_0\{(\hat{w}_\pi - w_\pi)(\hat{q}_{V_{k-1}} - q_{V_{k-1}})\}\|_{2,\rho} \right\} \end{aligned}$$

858 *Proof.* Equation 5 in the proof of Theorem 3 shows that, for any transformation $f \in \mathcal{G}_B$ and any k ,

$$\frac{1}{n} \sum_{i=1}^n f(\hat{v}(S_i)) \{\hat{Y}_{V_{k-1}}(O_i) - V_k(S_i)\} = 0.$$

859 Applying the above with $f \circ \hat{v} = V_B^{*(k)} - V_k$, where

$$V_B^{*(k)} := \mathcal{B}_{\pi,B}(V_{k-1}),$$

860 we obtain

$$\frac{1}{n} \sum_{i=1}^n (V_B^{*(k)} - V_k)(S_i) \{\hat{Y}_{V_{k-1}}(O_i) - V_k(S_i)\} = 0.$$

861 Adding and subtracting P_0 , we find

$$\begin{aligned} & P_0 \left[(V_B^{*(k)} - V_k) \{\hat{Y}_{V_{k-1}} - V_k\} \right] \\ & = (P_0 - P_n) \left[(V_B^{*(k)} - V_k) \{\hat{Y}_{V_{k-1}} - V_k\} \right]. \end{aligned} \tag{10}$$

862 First, we study the left-hand side of (10). We have

$$\begin{aligned} & P_0 \left[(V_B^{*(k)} - V_k) \{\hat{Y}_{V_{k-1}} - V_k\} \right] \\ & = \|V_B^{*(k)} - V_k\|_{2,\rho}^2 \\ & \quad + P_0 \left[(V_B^{*(k)} - V_k) \{\hat{Y}_{V_{k-1}} - V_B^{*(k)}\} \right]. \end{aligned}$$

863 Since $V_B^{*(k)} = \Pi_B \mathcal{T}_\pi V_{k-1}$ and $V_B^{*(k)} - V_k \in \mathcal{H}_B(\hat{v})$, the projection identity gives

$$P_0 \left[(V_B^{*(k)} - V_k) \{V_B^{*(k)} - \mathcal{T}_\pi V_{k-1}\} \right] = 0.$$

864 Thus

$$\begin{aligned} & P_0 \left[(V_B^{*(k)} - V_k) \{\hat{Y}_{V_{k-1}} - V_B^{*(k)}\} \right] \\ & = P_0 \left[(V_B^{*(k)} - V_k) \{\hat{Y}_{V_{k-1}} - \mathcal{T}_\pi V_{k-1}\} \right]. \end{aligned}$$

865 By Theorem 2 and arguing as in the proof of Theorem 3, it follows that

$$\begin{aligned} & \left| P_0 \left[(V_B^{*(k)} - V_k) \{\hat{Y}_{V_{k-1}} - V_B^{*(k)}\} \right] \right| \\ & \leq \|V_B^{*(k)} - V_k\|_{2,\rho} \|b_0\{(\hat{w}_\pi - w_\pi)(\hat{q}_{V_{k-1}} - q_{V_{k-1}})\}\|_{2,\rho}. \end{aligned}$$

866 Putting it all together,

$$\begin{aligned} & \|V_B^{*(k)} - V_k\|_{2,\rho}^2 \leq P_0 \left[(V_B^{*(k)} - V_k) \left\{ \widehat{Y}_{V_{k-1}} - V_k \right\} \right] \\ & + \|V_B^{*(k)} - V_k\|_{2,\rho} \|b_0\{(\widehat{w}_\pi - w_\pi)(\widehat{q}_{V_{k-1}} - q_{V_{k-1}})\}\|_{2,\rho}. \end{aligned}$$

867 Hence, by (10),

$$\begin{aligned} & \|V_B^{*(k)} - V_k\|_{2,\rho}^2 \\ & \leq (P_0 - P_n) \left[(V_B^{*(k)} - V_k) \left\{ \widehat{Y}_{V_{k-1}} - V_k \right\} \right] \\ & + \|V_B^{*(k)} - V_k\|_{2,\rho} \|b_0\{(\widehat{w}_\pi - w_\pi)(\widehat{q}_{V_{k-1}} - q_{V_{k-1}})\}\|_{2,\rho}. \end{aligned} \tag{11}$$

868 Next, we obtain a high-probability bound for the first term right-hand side of (11). Applying
869 Lemmas 3, 4, and 5, and arguing as in the proof of Theorem 3, we obtain that, with probability at
870 least $1 - \delta$,

$$\begin{aligned} & \left| (P_0 - P_n) \left[(V_B^{*(k)} - V_k) \left\{ \widehat{Y}_{V_{k-1}} - V_k \right\} \right] \right| \\ & \lesssim \delta_n^2 + \delta_n \|V_B^{*(k)} - V_k\|_{2,\rho} \\ & + \frac{\sqrt{\log(1/\delta)} \|V_B^{*(k)} - V_k\|_{2,\rho}}{\sqrt{n}} + \frac{\log(1/\delta)}{n}, \end{aligned}$$

871 where $\delta_n := \sqrt{(B/n) \log(n/B)}$, and the implicit constants do not depend on B .

872 Plugging this high probability bound into (11), we obtain that, with probability at least $1 - \delta$,

$$\begin{aligned} & \|V_B^{*(k)} - V_k\|_{2,\rho}^2 \\ & \lesssim \delta_n^2 + \delta_n \|V_B^{*(k)} - V_k\|_{2,\rho} \\ & + \frac{\sqrt{\log(1/\delta)} \|V_B^{*(k)} - V_k\|_{2,\rho}}{\sqrt{n}} + \frac{\log(1/\delta)}{n} \\ & + \|V_B^{*(k)} - V_k\|_{2,\rho} \|b_0\{(\widehat{w}_\pi - w_\pi)(\widehat{q}_{V_{k-1}} - q_{V_{k-1}})\}\|_{2,\rho}. \end{aligned}$$

873 Recalling that $\delta_n := \sqrt{\frac{B}{n} \log\left(\frac{n}{B}\right)}$, the inequality above implies that, with probability at least $1 - \delta$,

$$\begin{aligned} \|V_B^{*(k)} - V_k\|_{2,\rho} & \lesssim \sqrt{\frac{B}{n} \log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(1/\delta)}{n}} \\ & + \|b_0\{(\widehat{w}_\pi - w_\pi)(\widehat{q}_{V_{k-1}} - q_{V_{k-1}})\}\|_{2,\rho}. \end{aligned}$$

874 Taking the failure probability to be δ/K and applying a union bound over $k = 1, \dots, K$ proves the
875 result. $\square$

876 K.3 Proof of Theorem 4

877 *Proof of Theorem 4.* For $C < \infty$ large enough, define

$$\begin{aligned} \eta_k = C \Bigg\{ & \sqrt{\frac{B}{n} \log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(K/\delta)}{n}} \\ & + \|b_0\{(\widehat{w}_\pi - w_\pi)(\widehat{q}_{V_{k-1}} - q_{V_{k-1}})\}\|_{2,\rho} \Bigg\} \end{aligned}$$

878 By Lemma 9, with probability at least $1 - \delta$, uniformly over $k = 1, \dots, K$,

$$\|V_k - \mathcal{B}_{\pi,B}(V_{k-1})\|_{2,\rho} \leq \eta_k.$$

879 Lemma 8 therefore gives

$$\begin{aligned} \|V_K - V_B^*\|_{2,\rho} &\leq \gamma^K \|\hat{v} - V_B^*\|_\infty \\ &\quad + \frac{\sqrt{\mathcal{C}_{B,\gamma}}}{1-\gamma} \max_{1 \leq j \leq K} \eta_j. \end{aligned}$$

880 Combining this with Lemma 7 and the triangle inequality yields

$$\begin{aligned} \|V_K - v_0\|_{2,\rho} &\leq \gamma^K \|\hat{v} - V_B^*\|_\infty \\ &\quad + \frac{\sqrt{\mathcal{C}_{B,\gamma}}}{1-\gamma} \left[\|(I - \Pi_B)v_0\|_{2,\rho} + \max_{1 \leq j \leq K} \eta_j \right]. \end{aligned}$$

881 Since Π_B is the $L^2(\rho)$ projection onto $\mathcal{H}_B(\hat{v})$,

$$\|(I - \Pi_B)v_0\|_{2,\rho} = \min_{h \in \mathcal{H}_B(\hat{v})} \|h - v_0\|_{2,\rho}.$$

882 Substituting the definition of η_k and absorbing constants completes the proof. $\square$

883 K.4 Bound on Successive Iteration Errors

884 **Lemma 10** (Bound on Successive Iteration Errors). *Under the conditions of Theorem 4,*

$$\begin{aligned} &\|V_K - V_{K-1}\|_{2,\rho} \\ &\lesssim \gamma^{K-1} \|\hat{v} - V_B^*\|_\infty \\ &\quad + \frac{\sqrt{\mathcal{C}_{B,\gamma}}}{1-\gamma} C \left(\sqrt{\frac{B}{n} \log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(K/\delta)}{n}} \right. \\ &\quad \left. + \max_{1 \leq j \leq K} \left\| b_0\{(\hat{w}_\pi - w_\pi)(\hat{q}_{V_{j-1}} - q_{V_{j-1}})\} \right\|_{2,\rho} \right). \end{aligned}$$

885 *Proof.* Lemma 8 and Lemma 9 give the same bound for $\|V_m - V_B^*\|_{2,\rho}$, for $m = K-1$ and $m = K$,
886 with $\gamma^m \|\hat{v} - V_B^*\|_\infty$ as the finite-iteration term. By the triangle inequality,

$$\begin{aligned} \|V_K - V_{K-1}\|_{2,\rho} &\leq \|V_{K-1} - V_B^*\|_{2,\rho} \\ &\quad + \|V_K - V_B^*\|_{2,\rho}. \end{aligned}$$

887 Substituting the two bounds and absorbing constants proves the claim. $\square$

888 L Proofs of Theorems 5 and 6

889 *Proof of Theorem 5.* The first-order optimality conditions for isotonic regression (equivalently, its
890 interpretation as a histogram estimator) imply that for any function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is a linear
891 combination of the indicator functions defining the isotonic partition,

$$\frac{1}{n} \sum_{i=1}^n f(V_K(S_i)) \left\{ \hat{Y}_{V_{K-1}}(O_i) - V_K(S_i) \right\} = 0.$$

892 See, for example, the proof of Lemma C.1 in Van Der Laan et al. [60] and van der Laan et al. [61].
893 Choosing f appropriately yields

$$\begin{aligned} 0 &= \frac{1}{n} \sum_{i=1}^n \left\{ \Gamma_0(V_K)(S_i) - V_K(S_i) \right\} \\ &\quad \times \left\{ \hat{Y}_{V_{K-1}}(O_i) - V_K(S_i) \right\}, \end{aligned}$$

894 which is the same basic equality as (6) in the proof of Theorem 3. The remainder of the argument
895 proceeds along the same lines with minor modifications.

Specifically, let $\mathcal{F}_{TV}$ denote the union of $\mathcal{G}_{\text{iso}}$, which is uniformly bounded by $2M$ under Condition C1, with all functions of bounded total variation bounded by the constant C in Condition C5. By Van Der Vaart and Wellner [64], this class satisfies the uniform entropy integral bound $\mathcal{J}(\delta, \mathcal{F}_{TV}) \lesssim \sqrt{\delta}$.

Let

$$m_K(t) := E[(\mathcal{T}_\pi V_K)(S) \mid \hat{v}(S) = t, \mathcal{C}_n].$$

Condition C5 states that m_K has bounded total variation. Since

$$\Gamma_0(V_K)(S) = E\{m_K(\hat{v}(S)) \mid V_K(S)\},$$

and V_K is a nondecreasing function of $\hat{v}$, Lemma 6 of van der Laan et al. [61] implies that the coarsened curve $\Gamma_0(V_K)$ also has bounded total variation and lies in $\mathcal{F}_{TV, \hat{v}} := \{f \circ \hat{v} : f \in \mathcal{F}_{TV}\}$. Thus, it holds that

$$(\Gamma_0(V_K) - V_K)(\hat{Y}_{V_{K-1}} - V_K)$$

lies in a uniformly bounded subset of the class

$$\hat{\mathcal{G}} := \left\{ (f_1 - f_2)(\hat{Y}_{f_2} - f_2) : f_1, f_2 \in \mathcal{F}_{TV, \hat{v}} \right\}.$$

Arguing as in the proof of Theorem 3, we have

$$\mathcal{J}(\delta, \hat{\mathcal{G}}) \lesssim \mathcal{J}(\delta, \mathcal{F}_{TV}) \lesssim \sqrt{\delta}.$$

The proof now follows directly from Theorem 3 with this new choice of $\hat{\mathcal{G}}$ and its associated critical radius $\delta_n = n^{-1/3}$ for monotone functions. $\square$

Proof of Theorem 6. The result follows directly from a union bound and the proofs of Theorems 3 and 4, with $\mathcal{G}_B$ replaced by $\mathcal{G}_{B_n}$, the class of all piecewise-constant functions with at most B_n constant segments. In particular, the entropy bound in Lemma 5 continues to apply to this class. Notably, the proofs of these theorems allow for data-adaptive partitions and require only a deterministic upper bound on the number of constant segments. $\square$

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